The Astrophysical Journal, 898:83 (19pp), 2020 July 20 https://doi.org/10.3847/1538-4357/ab9dff
© 2020. The American Astronomical Society. All rights reserved.

The Black Hole Mass–Color Relations for Early- and Late-type Galaxies: Red and Blue Sequences

Bililign T. Dullo[1], Alexandre Y. K. Bouquin[1,2,3], Armando Gil de Paz[1], Johan H. Knapen[2,3], and Javier Gorgas[1]
[1] Departamento de Física de la Tierra y Astrofísica, Instituto de Física de Partículas y del Cosmos IPARCOS, Universidad Complutense de Madrid, E-28040 Madrid, Spain; bdullo@ucm.es
[2] Instituto de Astrofísica de Canarias, Vía Láctea S/N, E-38205 La Laguna, Spain
[3] Departamento de Astrofísica, Universidad de La Laguna, E-38206 La Laguna, Spain
Received 2019 December 26; revised 2020 May 25; accepted 2020 June 13; published 2020 July 27

Abstract

The tight correlations between supermassive black hole (SMBH) mass ($M_{\rm BH}$) and the properties of the host galaxy have useful implications for our understanding of the growth of SMBHs and of the evolution of galaxies. Here, we present newly observed correlations between $M_{\rm BH}$ and the host galaxy total UV−[3.6] color ($\mathcal{C}_{\rm UV,tot}$, Pearson's $r = 0.6$–0.7) for a sample of 67 galaxies (20 early-type galaxies and 47 late-type galaxies) with directly measured $M_{\rm BH}$ in the Galaxy Evolution Explorer/S^{4}G survey. The colors are carefully measured in a homogeneous manner using the far-UV, near-UV, and 3.6 μm magnitudes of the galaxies and their multicomponent structural decompositions in the literature. We find that more massive SMBHs are hosted by (early- and late-type) galaxies with redder colors, but the $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ relations for the two morphological types have slopes that differ at $\sim 2\sigma$ level. Early-type galaxies define a red sequence in the $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ diagrams, while late-type galaxies trace a blue sequence. Within the assumption that the specific star formation rate of a galaxy (sSFR) is well traced by $L_{\rm UV}/L_{3.6}$, it follows that the SMBH masses for late-type galaxies exhibit a steeper dependence on sSFR than those for early-type galaxies. The $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ and $M_{\rm BH} - L_{3.6,\rm tot}$ relations for the sample galaxies reveal a comparable level of vertical scatter in the log $M_{\rm BH}$ direction, approximately 5%–27% more than the vertical scatter of the $M_{\rm BH} - \sigma$ relation. Our $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ relations suggest different channels of SMBH growth for early- and late-type galaxies, consistent with their distinct formation and evolution scenarios. These new relations offer the prospect of estimating SMBH masses reliably using the galaxy color alone. Furthermore, we show that they are capable of estimating intermediate black hole masses in low-mass early- and late-type galaxies.

Unified Astronomy Thesaurus concepts: Supermassive black holes (1663); cD galaxies (209); Elliptical galaxies (456); Lenticular galaxies (915); Spiral galaxies (1560); Galaxy colors (586); Galaxy photometry (611); Galaxy nuclei (609); Galaxy structure (622)

Supporting material: machine-readable tables

1. Introduction

Almost all local galaxies are believed to harbor a supermassive black hole (SMBH, $M_{\rm BH} \sim 10^5$–$10^9 M_\odot$) at their center (Magorrian et al. 1998; Richstone et al. 1998; Ferrarese & Ford 2005). The connection between SMBHs and their host galaxies has been a subject of interest since Kormendy & Richstone (1995, see also Dressler 1989) first reported a linear correlation between SMBH mass ($M_{\rm BH}$) and the luminosity of the host bulge (i.e., the entire galaxy in case of elliptical galaxies), see Kormendy & Ho (2013) and Graham (2016) for recent reviews. SMBH masses scale with a number of host galaxy properties, including stellar velocity dispersion (σ, Ferrarese & Merritt 2000; Gebhardt et al. 2000), bulge luminosity ($L_{\rm bulge}$) and bulge mass ($M_{\rm bulge}$, Kormendy & Richstone 1995; Magorrian et al. 1998), a depleted stellar core (e.g., Dullo 2019), and stellar concentration (Graham et al. 2001). Not only did these scaling relations allow us to predict $M_{\rm BH}$ in galaxies, but they also led to a suggestion that the fueling and growth rate of the central SMBH are intimately coupled to the star formation rate and stellar mass build-up of the host galaxy. However, the exact nature of the physical mechanism driving this connection remains unclear.

Accretion of gas onto SMBHs triggers active galactic nucleus (AGN) feedback, which is critical for the regulation of the star formation and growth of the host galaxy (e.g., Silk & Rees 1998; Fabian 1999; Di Matteo et al. 2005; Springel et al. 2005; Croton et al. 2006; Hopkins et al. 2006). The distinct formation histories, colors, and structural properties of early- and late-type galaxies may therefore reflect two different channels of gas accretion and SMBH growth for the two morphological types (see Hopkins & Hernquist 2009; Shankar et al. 2009; Schawinski et al. 2010, 2014; Krajnović et al. 2018). Observations show that early-type (i.e., lenticular and elliptical) galaxies do not correlate with SMBHs in the same manner as late-type (i.e., spiral) galaxies do (e.g., McConnell & Ma 2013; Saglia et al. 2016; Savorgnan & Graham 2016; Krajnović et al. 2018; Sahu et al. 2019a, 2019b). Early-type galaxies are typically red with a low level of star formation rate, while late-type galaxies have blue colors and are actively star-forming. This dichotomy gives rise to a bimodal distribution of galaxies in the color–magnitude diagram—early-type galaxies define a red sequence that is separate from late-type galaxies, which reside in a blue cloud (e.g., Chester & Roberts 1964; Visvanathan & Griersmith 1977; Baldry et al. 2004; Brammer et al. 2009).

In the hierarchical galaxy formation scenario, early-type galaxies are built through mergers of smaller systems and accretion events (e.g., White & Rees 1978; Naab et al. 2006b; Hopkins et al. 2009a, 2009b; Rodriguez-Gomez et al. 2016; Mundy et al. 2017). Major-merger-driven inflow of gas into the nuclear regions of the newly formed merger remnant can produce rapid bursts of star formation and fuel the SMBH

(Barnes & Hernquist 1991, 1996; Naab et al. 2006a; Hopkins & Quataert 2010). For late-type (i.e., spiral) galaxies, one of the most frequently advocated formation scenarios is secular evolution involving nonaxisymmetric stellar structures, such as bars and spiral arms, which can drive an inflow of gas from the disk into the nuclear region and onto the central SMBH (Kormendy 1982; Courteau et al. 1996; Carollo et al. 1997; Kormendy & Kennicutt 2004; Athanassoula 2005; Laurikainen et al. 2007; Fisher & Drory 2008; Graham & Worley 2008; Gadotti 2009; Dullo et al. 2016, 2019; Tonini et al. 2016).

A related issue is the observed departures from single power-law relations (e.g., bends and offsets) in SMBH scaling diagrams at the low-mass and high-mass ends or when the galaxy sample contains Sérsic and core-Sérsic galaxies. In particular, Graham (2012, see also Graham & Scott 2013) reported two separate $M_{\rm BH} - L_{\rm bulge}$ relations with distinct slopes for Sérsic and core-Sérsic galaxies. Core-Sérsic galaxies are luminous ($M_B \lesssim -20.5 \pm 0.5$ mag) galaxies that exhibit a flattening in their inner stellar light distribution due to a central deficit of light relative to the inward extrapolation of their outer Sérsic (1968) light profile (Faber et al. 1997; Graham et al. 2003; Hopkins et al. 2009b; Dullo & Graham 2012, 2013, 2014, 2015; Dullo et al. 2017, 2018; Dullo 2019). They are thought to have formed from a few gas-poor major merger events,[4] but a small fraction of them can host molecular gas reservoirs that feed an ongoing low level of star formation (Davis et al. 2019). In contrast, the low- and intermediate luminosity ($M_B \gtrsim -20.5$ mag) Sérsic galaxies, with no depleted cores, are the product of gas-rich mergers (Hopkins et al. 2009a; Dullo & Graham 2012, 2014; Dullo et al. 2016, 2019). Departures from the best-fitting single power-law relation in a black hole scaling diagram may hold implications for SMBH and galaxy coevolution, but because the majority of SMBH scaling relations to date are based on host galaxy properties that solely trace the old stellar populations and disregard the young and intermediate-age stars, the exact mechanism to establish the assumed coupling between the growth of the SMBHs and the build-up of their host galaxies is unclear.

Here, we explore a scenario in which the complex interplay between the details of the SMBH growth, efficiency of AGN feedback, regulated star formation histories and major merger histories of the host galaxy establish a relation between the SMBH and color of the host galaxy (see also Schawinski et al. 2010, 2014). Emerging evidence has shown a link between $M_{\rm BH}$ and the star formation rate (SFR) in nearby galaxies. Terrazas et al. (2017) reported an inverse correlation between the specific star formation rate (sSFR) and specific SMBH mass. Martín-Navarro et al. (2018) observed a trend between SMBH mass and host galaxy star formation histories, where the star formation in galaxies hosting more massive SMBHs was quenched early and more efficiently than in those with less massive SMBHs (see also van Son et al. 2019). Because the UV−[3.6] color ($\mathcal{C}_{\rm UV}$) is a good proxy for the sSFR (e.g., Bouquin et al. 2018), a correlation between $M_{\rm BH}$ and UV−[3.6] color is expected. The UV flux is a proxy for the current SFR because it traces massive young stars. The emission at 3.6 μm, which is less affected by dust extinction, is a good proxy for stellar mass because it primarily traces older stellar populations. However, a small fraction (5%–15%) of the 3.6 μm flux can be due to contributions from intermediate-age stars, polycyclic aromatic hydrocarbons, and hot dust (Meidt et al. 2012). The $M_{\rm BH} -$ color relation has so far been overlooked, in part owing to the narrow wavelength baselines that are commonly used to determine colors.

In this paper, we present for the first time correlations between $M_{\rm BH}$ and $\mathcal{C}_{\rm UV}$ colors for 67 Galaxy Evolution Explorer (GALEX)/S^{4}G galaxies with directly measured SMBH masses (van den Bosch 2016) and homogeneously determined GALEX far-UV (FUV), near-UV (NUV), and Spitzer 3.6 μm magnitudes (Bouquin et al. 2018). By dividing the sample galaxies by morphology, we fit two different $M_{\rm BH} - \mathcal{C}_{\rm UV}$ relations with distinct slopes for early- and late-type galaxies.

The $M_{\rm BH} - \mathcal{C}_{\rm UV}$ relation has multiple important applications. It allows the SMBHs in early- and late-type galaxies to be predicted free from uncertainties due to distances and mass-to-light ratios, although there are uncertainties due to K-corrections for more distant galaxies. Furthermore, using the $M_{\rm BH} - \mathcal{C}_{\rm UV}$ relation, we hope to properly understand the poorly constrained low-mass end of the SMBH scaling relations. In so doing, we can predict SMBHs or intermediate-mass black holes (IMBHs) with masses $\sim$100–$10^5 M_\odot$ in low-mass systems and bulgeless spiral galaxies. In addition, because colors are easy to measure even for high-redshift galaxies, studying the galaxy color and black hole mass evolution at different epochs may provide further clues on the different channels of black hole growth.

The paper is organized as follows. Section 2.1 describes the sample selection. Sections 2.2 and 2.3 describe the SMBH data, and the UV and 3.6 μm apparent asymptotic magnitudes. The derivation of bulge, disk, and total magnitudes for the sample galaxies along with the corresponding dust corrections and error measurements are discussed in Sections 3.1 and 3.2, respectively. We proceed to discuss the regression techniques we employed to fit the black hole scaling relations in Section 3.3 and present the results from our regression analyses in Sections 3.4 and 3.5. Section 4 provides a discussion of our results, including the origin and implications of the $M_{\rm BH} - \mathcal{C}_{\rm UV}$ relations. Section 5 summarizes our main conclusions.

There are four appendices at the end of this paper (Appendices A–D). Appendix A discusses our implementation of the Markov chain Monte Carlo (MCMC) Bayesian statistical method. Notes on five notable outliers in the $M_{\rm BH} - \mathcal{C}_{\rm UV}$ relations are given in Appendix B. Appendix C includes a table that lists apparent total magnitudes, flux ratios, dust corrections, and direct SMBH masses for our sample galaxies. We tentatively predict black hole masses in a sample of 1382 GALEX/S^{4}G galaxies with no measured black hole masses using our $M_{\rm BH} - \mathcal{C}_{\rm UV}$ relations and tabulate them in Appendix D.

2. Sample and Data

2.1. Sample Selection

Our investigation of correlations between the SMBH mass ($M_{\rm BH}$) and (UV−[3.6]) galaxy color uses UV and 3.6 μm magnitudes that are determined in a homogeneous manner for a large sample of galaxies with measured SMBH masses. Henceforth, we refer to the (UV−[3.6]) color as $\mathcal{C}$. Bouquin et al. (2015, 2018) provided FUV ($\lambda_{\rm eff} \sim 1526$ Å), NUV ($\lambda_{\rm eff} \sim 2267$ Å), and 3.6 μm asymptotic magnitudes for a sample of 1931 nearby galaxies taken from the Spitzer Survey

[4] The depleted cores of core-Sérsic galaxies are thought to be scoured by coalescing binary SMBHs formed in the gas-poor merger events (Begelman et al. 1980; Ebisuzaki et al. 1991; Merritt 2006).

of Stellar Structure in Galaxies (S^4G) sample (Sheth et al. 2010). They used UV and near-infrared (3.6 μm) imaging data obtained with the GALEX, (Martin et al. 2005; Gil de Paz et al. 2007) and the Infrared Array Camera on the Spitzer Space Telescope, respectively. We use the SMBH sample presented in van den Bosch (2016, his Table 2). They publish a compilation of directly measured SMBH masses ($M_{\rm BH}$), half-light radii ($R_{\rm e}$), and total K_s-band luminosities (L_k) for a large sample of 245 galaxies. We selected all galaxies that were in common with Bouquin et al. (2018) and van den Bosch (2016), resulting in a sample of 67 galaxies studied in this paper. Homogenized mean central velocity dispersions (σ) and morphological classifications (elliptical E, elliptical-lenticular E-S0, lenticular S0, lenticular-spiral S0-a, and spiral S) of the sample galaxies were obtained from Hyperleda[5] (Paturel et al. 2003; Makarov et al. 2014). In the analysis of the scaling relations (Tables 1–4), the galaxies are divided into two broad morphological classes, early-type galaxies (9 Es, 2 E-S0s, 2 S0s, and 7 S0-a) and late-type galaxies (47 Ss). Most of our late-type galaxies are disk dominated. The full list of galaxies and their properties are presented in Appendix C.

In Figure 1 we determine potential sample selection biases by comparing $M_{\rm BH}$, $R_{\rm e}$, and L_k of our sample and those of another 178 known galaxies with measured black hole masses using data from van den Bosch (2016, his Table 2). Both the early- and late-type galaxies in our sample probe a wide range in $M_{\rm BH}$, $R_{\rm e}$, and L_k to allow a robust investigation of the $M_{\rm BH} - \mathcal{C}$ relations. We also find that our early- and late-type galaxies span a wide range in stellar mass ($M_{*,k}$), Figure 1(c). We compute the galaxy stellar masses using L_k and assuming the K_s-band mass-to-light ratio $M_*/L_k = 0.10\sigma^{0.45}$ (van den Bosch 2016). Late-type galaxies in our sample have $M_{*,k}$ that range from $10^9 M_\odot$ to $2 \times 10^{11} M_\odot$, and for 75% of them, $M_{*,k}/M_\odot \gtrsim 2 \times 10^{10}$. For the early-type galaxies, $10^{10} \lesssim M_{*,k}/M_\odot \lesssim 10^{12}$, and 90% of these galaxies have $M_{*,k}/M_\odot \gtrsim 2 \times 10^{10}$. It worth noting that the GALEX/S^4G sample galaxies (Bouquin et al. 2018) were chosen to have radio-derived radial velocities of $V_{\rm radio} < 3000$ Km s^{-1} in HyperLEDA. The sample therefore lacks H I-poor extremely massive galaxies, including brightest cluster galaxies (BCGs).

2.2. Black Hole Masses

Our selected sample of 67 galaxies includes 64 galaxies with SMBH masses ($M_{\rm BH}$) that are determined from stellar, gas, or maser kinematic measurements. For the remaining 3 galaxies (NGC 4051, NGC 4593, and NGC 5273), $M_{\rm BH}$ measurements were based on reverberation mapping (Bahcall et al. 1972; Peterson 1993). Of the 67 galaxies, 62 harbor black holes that are supermassive (i.e., $M_{\rm BH} \gtrsim 10^6 M_\odot$); see Appendix C. While the remaining 5 sample galaxies have $2 \times 10^4 M_\odot \lesssim (M_{\rm BH}) \lesssim 3 \times 10^5 M_\odot$, we also refer to these black holes as SMBHs. The sample includes 25 galaxies with $M_{\rm BH}$ upper limits, comprised of 23 late-type galaxies and 2 early-type galaxies. We discuss the effect of including upper limits on the black hole scaling relations in Section 3. The uncertainties on $M_{\rm BH}$ were taken from van den Bosch (2016).

[5] http://leda.univ-lyon1.fr

2.3. UV and 3.6 μm Apparent Asymptotic Magnitudes

Bouquin et al. (2015, 2018) used the 3.6 μm surface brightness profiles from Muñoz-Mateos et al. (2015) to determine the 3.6 μm magnitudes for their galaxies. They extracted the UV surface brightness profiles of the galaxies following the prescription of Muñoz-Mateos et al. (2015). Briefly, the sky levels in galaxy images were first determined, then the images were masked to avoid bright foreground and background objects (Bouquin et al. 2018, their Section 3.1). The FUV, NUV, and 3.6 μm radial surface brightness profiles were then extracted using a series of elliptical annuli with fixed ellipticity and position angle, after the innermost regions, $R < 3''$, were excluded (Bouquin et al. 2015, 2018; Muñoz-Mateos et al. 2015). Each annulus has a width of $6''$, and measurements were taken up to $3\times$ the major axis of the galaxy D25 elliptical isophote. To derive the FUV, NUV, and 3.6 μm asymptotic magnitudes from the corresponding surface brightness profiles, Bouquin et al. (2018) extrapolated the growth curves. The UV and 3.6 μm data reach depths of $\sim$27 mag arcsec^{-2} and $\sim$26.5 mag arcsec^{-2}, respectively, allowing the growth curves to flatten out for most galaxies. The method yielded robust asymptotic magnitudes with errors that are within the FUV and NUV zero-point uncertainties of 0.05 and 0.03 AB mag (Morrissey et al. 2007).

3. Results

3.1. Calculating Dust-corrected Total Magnitudes

The FUV, NUV, and 3.6 μm (total) asymptotic magnitudes of our sample were corrected for Galactic extinction by Bouquin et al. (2015, 2018) using the $E(B-V)$ reddening values taken from Schlegel et al. (1998). For the analyses in the paper, we define the total galaxy luminosity as the sum of the luminosities of the bulge and disk, excluding additional galaxy structural components such as bars and rings. The bulk of the galaxies in our sample are multicomponent systems. Fractional luminosities of the individual structural components are needed to obtain accurate bulge and disk magnitudes from the total asymptotic magnitudes. We note that from here on, the term "bulge" is used when we refer to the spheroids of elliptical galaxies and to the bulges of disk galaxies. Fortunately, Salo et al. (2015) performed detailed 2D multicomponent decompositions of 3.6 μm images of all our galaxies into nuclear sources, bulges, disks, and bars, and they presented fractional bulge and disk luminosities that we use to calculate the 3.6 μm bulge and disk magnitudes. For two galaxies (NGC 3368 and NGC 4258) with poor fits in Salo et al. (2015), the 3.6 μm fractional luminosities were taken from Savorgnan & Graham (2016). The bulge-to-disk flux ratio (B/D) of a galaxy depends on the observed wavelength and galaxy morphological type (e.g., Möllenhoff 2004; Möllenhoff et al. 2006; Graham & Worley 2008; Kennedy et al. 2016). The UV B/T and D/T tabulated in Appendix C were derived using the 3.6 μm B/D together with the (extrapolation of the) B/D passband-morphological type diagram from Möllenhoff (2004, his Figure 5).

For the disk galaxies, we additionally applied the inclination-dependent (i) internal dust attenuation corrections from Driver et al. (2008, their Equations (1) and (2) and Table 1) to determine dust-corrected bulge and disk magnitudes. Because Driver et al. (2008) did not provide 3.6 μm dust corrections, we rely on the prescription for their reddest bandpass (i.e., K) to

Table 1
$M_{\rm BH} - \mathcal{C}_{\rm FUV,tot}$ Scaling Relations for Early- and Late-type Galaxies

			$Y = \beta X + \alpha$, $X = \mathcal{C}_{\rm FUV,tot} - 6.5$, $Y = \log M_{\rm BH}$							
			early-type galaxies, Figure 2, left							
Regression method	$Y\|X$		$X\|Y$		Bisector		r	Δ (dex)	ϵ (dex)	N
	α	β	α	β	α	β				
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
BCES	8.12 ± 0.18	1.28 ± 0.40	8.30 ± 0.31	2.65 ± 0.82	**8.18 ± 0.20**	**1.75 ± 0.41**	0.61	0.86	⋯	18
LINMIX_ERR	8.15 ± 0.22	1.25 ± 0.44	8.34 ± 0.23	2.47 ± 0.84	⋯	1.71 ± 0.51	⋯	⋯	0.69 ± 0.20	18
OLS	8.10 ± 0.17	1.04 ± 0.30	8.32 ± 0.27	2.59 ± 0.77	8.17 ± 0.18	1.57 ± 0.27	⋯	0.87	⋯	18
					Symmetric					
MCMC	⋯	⋯	⋯	⋯	8.27 ± 0.25	2.06 ± 0.63	⋯	0.85	⋯	18
			$Y = \beta X + \alpha$, $X = \mathcal{C}_{\rm FUV,tot} - 3.3$, $Y = \log M_{\rm BH}$							
			late-type galaxies, Figure 2, left							
	$Y\|X$		$X\|Y$		Bisector		r	Δ (dex)	ϵ (dex)	N
	α	β	α	β	α	β				
BCES	7.01 ± 0.13	0.82 ± 0.20	7.06 ± 0.19	1.35 ± 0.36	**7.03 ± 0.14**	**1.03 ± 0.13**	0.60	0.87	⋯	45
LINMIX_ERR	7.07 ± 0.13	0.87 ± 0.26	7.09 ± 0.13	1.59 ± 0.38	⋯	1.17 ± 0.27	⋯	⋯	0.69 ± 0.22	45
OLS	6.97 ± 0.12	0.62 ± 0.16	7.10 ± 0.22	1.82 ± 0.36	7.02 ± 0.14	1.05 ± 0.12	⋯	0.87	⋯	45
					Symmetric					
MCMC	⋯	⋯	⋯	⋯	6.95 ± 0.11	0.87 ± 0.17	⋯	0.81	⋯	45
			$Y = \beta X + \alpha$, $X = \mathcal{C}_{\rm NUV,tot} - 5.0$, $Y = \log M_{\rm BH}$							
			early-type galaxies, Figure 2, right							
	$Y\|X$		$X\|Y$		Bisector		r	Δ (dex)	ϵ (dex)	N
	α	β	α	β	α	β				
BCES	8.07 ± 0.15	1.57 ± 0.36	8.21 ± 0.29	2.74 ± 0.99	**8.12 ± 0.17**	**1.95 ± 0.28**	0.70	0.72	⋯	19
LINMIX_ERR	8.10 ± 0.20	1.66 ± 0.55	8.19 ± 0.20	2.63 ± 0.79	⋯	2.04 ± 0.60	⋯	⋯	0.66 ± 0.18	19
OLS	8.06 ± 0.15	1.27 ± 0.28	8.18 ± 0.24	2.65 ± 0.68	8.11 ± 0.16	1.76 ± 0.20	⋯	0.68	⋯	19
					Symmetric					
MCMC	⋯	⋯	⋯	⋯	8.06 ± 0.26	2.33 ± 0.74	⋯	0.82	⋯	19
			$Y = \beta X + \alpha$, $X = \mathcal{C}_{\rm NUV,tot} - 2.7$, $Y = \log M_{\rm BH}$							
			late-type galaxies, Figure 2, right							
	$Y\|X$		$X\|Y$		Bisector		r	Δ (dex)	ϵ (dex)	N
	α	β	α	β	α	β				
BCES	6.96 ± 0.15	1.24 ± 0.35	6.96 ± 0.17	1.61 ± 0.33	**6.96 ± 0.15**	**1.38 ± 0.23**	0.65	0.86	⋯	45
LINMIX_ERR	7.02 ± 0.12	1.08 ± 0.30	8.32 ± 0.27	1.86 ± 0.41	⋯	⋯	⋯	⋯	0.66 ± 0.18	45
OLS	6.94 ± 0.12	0.82 ± 0.17	6.94 ± 0.18	1.94 ± 0.35	6.94 ± 0.13	1.24 ± 0.13	⋯	0.83	⋯	45
					Symmetric					
MCMC	⋯	⋯	⋯	⋯	6.90 ± 0.13	1.35 ± 0.26	⋯	0.85	⋯	45

Note. Correlation between SMBH mass ($M_{\rm BH}$) and total UV−[3.6] color ($\mathcal{C}_{\rm UV,tot}$) for our early- and late-type galaxies with directly measured $M_{\rm BH}$. $\mathcal{C}_{\rm FUV,tot} = m_{\rm FUV} - m_{3.6\mu m}$, $\mathcal{C}_{\rm NUV,tot} = m_{\rm NUV} - m_{3.6\mu m}$, where $m_{\rm FUV}$, $m_{\rm NUV}$ and $m_{3.6\mu m}$ are the FUV, NUV, and $m_{3.6\mu m}$ total apparent magnitudes of the galaxies (Table C1). Column (1) lists the regression method. Columns (2) and (3) are the intercepts (α) and slopes (β) from the ($Y|X$) regressions. Columns (4) and (5) are α and β obtained from the $Y|X$ regression fits, while Columns (6) and (7) show α and β from the symmetrical bisector regressions. The preferred slopes and intercepts are highlighted in bold. Column (8) lists the Pearson correlation coefficient (r). Column (9) lists the rms scatter around the fitted BCES bisector relation in the log $M_{\rm BH}$ direction (Δ). Column (10) lists the intrinsic scatter (ϵ), see the text for details. Column (11) lists the number of data points contributing to the regression fits.

correct the 3.6 μm magnitudes. These corrections are given by

$$m^{\rm corr}_{\rm bulge,UV} = m^{\rm obs}_{\rm bulge,UV} - 1.10 - 0.95[1 - \cos(i)]^{2.18}\,, \quad (1)$$

$$m^{\rm corr}_{\rm disk,UV} = m^{\rm obs}_{\rm disk,UV} - 0.45 - 2.31[1 - \cos(i)]^{3.42}\,, \quad (2)$$

$$m^{\rm corr}_{\rm bulge,3.6} = m^{\rm obs}_{\rm bulge,3.6} - 0.11 - 0.79[1 - \cos(i)]^{2.77}\,, \quad (3)$$

$$m^{\rm corr}_{\rm disk,3.6} = m^{\rm obs}_{\rm disk,3.6} - 0.04 - 0.46[1 - \cos(i)]^{4.23}\,, \quad (4)$$

where $\cos(i) = b/a$, i.e., the minor-to-major axis ratios, which were computed for our galaxies from the minor and major galaxy diameters given in the NED database.

For each sample galaxy, we derived the dust-corrected total galaxy magnitude as

$$m^{\rm corr}_{\rm total} = -2.5\log(10^{-0.4m^{\rm corr}_{\rm disk}} + 10^{-0.4m^{\rm corr}_{\rm bulge}}). \quad (5)$$

3.2. Uncertainties on Magnitudes

Analyses of the black hole–color relations can be affected by uncertainties on the bulge, disk, and total magnitudes (Section 3.1 and Appendix C), which are dominated by systematic errors. We account for four potential sources of systematic uncertainty. Uncertainties on the UV and 3.6 μm asymptotic magnitudes are introduced by dust contamination, imperfect sky background determination, and poor masking of

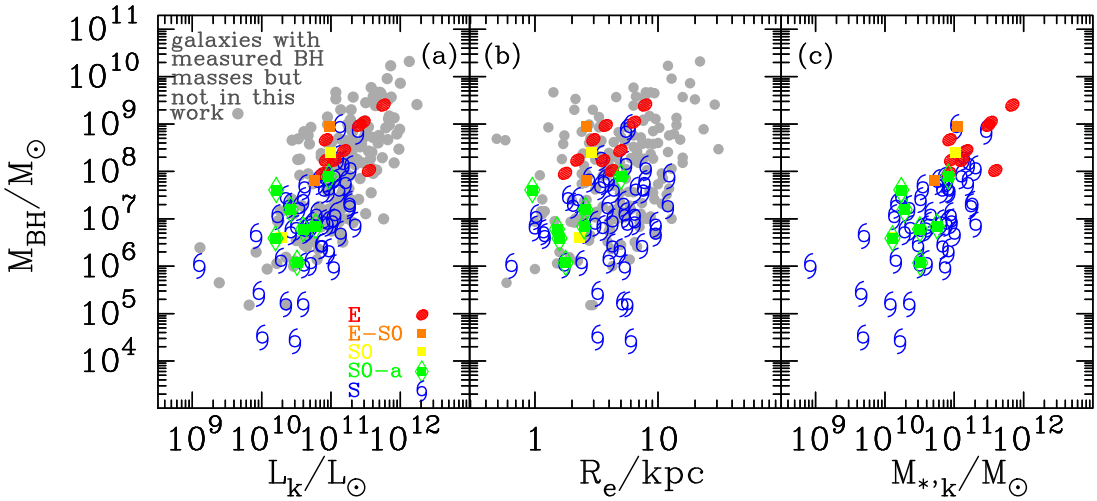

Figure 1. Galaxies with measured SMBH masses (M_{BH}). M_{BH} plotted as a function of (a) total K_s-band luminosity and (b) the half-light radius (R_e) for a sample of 245 galaxies with measured M_{BH} (van den Bosch 2016, his Table 2). Colored symbols denote our sample of 67 galaxies, whereas filled gray circles show the remaining 178 galaxies with measured M_{BH} in van den Bosch (2016, his Table 2). Panel (c): M_{BH} vs. total K_s-band stellar mass ($M_{*,k}$) for our sample derived from the total K_s-band luminosities assuming the K_s-band mass-to-light ratio $M_*/L_k = 0.10\sigma^{0.45}$ (van den Bosch 2016). Morphological classifications are from HyperLeda.

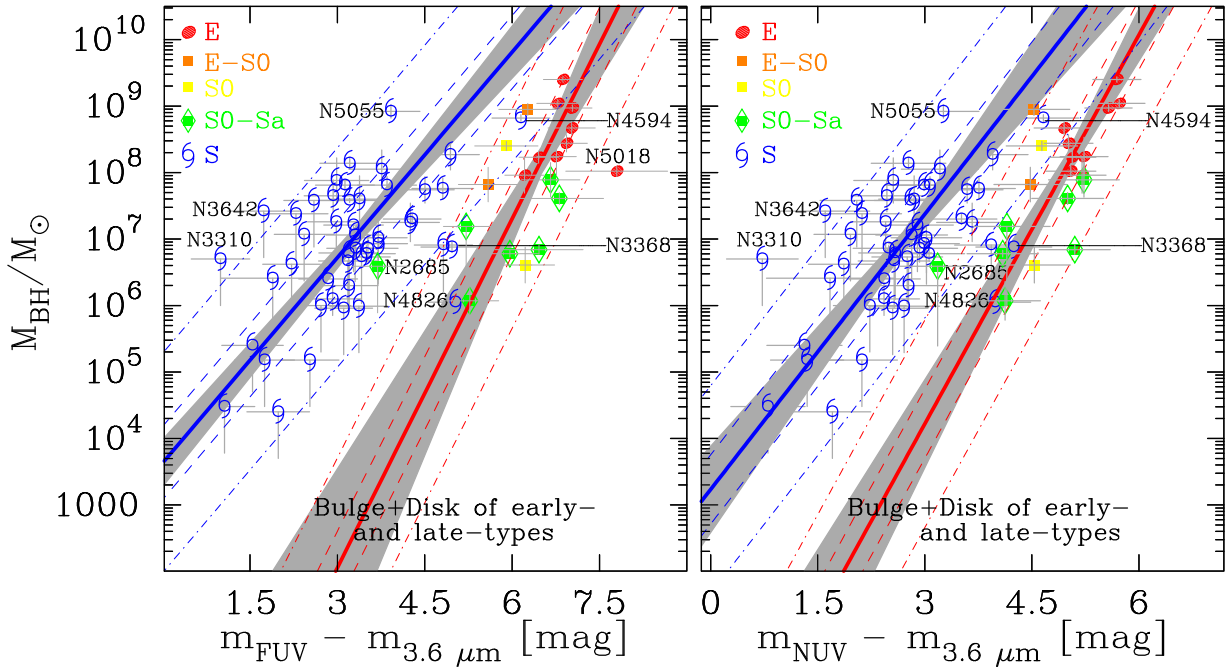

Figure 2. Correlations of directly measured SMBH masses (M_{BH}) with the total (i.e., bulge+disk) UV−3.6 μm colors ($\mathcal{C}_{UV,tot}$) of their host galaxies. $M_{BH} - \mathcal{C}_{FUV,tot}$ relations (left) and $M_{BH} - \mathcal{C}_{NUV,tot}$ relations (right) for early- and late-type galaxies. Our early-type morphological bin comprises E, E-S0, S0, and S0-a. Late-type galaxies (i.e., Sa, Sb, Sc, Sd, Sm, and Irr) are plotted in blue. Early- and late-type galaxies, which are fit separately, define two distinct red and blue sequences with significantly different slopes (see Table 1). The solid red and solid blue lines are the symmetric BCES bisector fits to our early- and late-type data, respectively. The shaded regions cover the associated 1σ uncertainties on these fits (Table 1). The dashed and dash–dotted lines delineate one and three times the intrinsic scatter, respectively. Errors on M_{BH} and $\mathcal{C}_{UV,tot}$ are shown, and for galaxies with black hole upper limits, we only show the lower uncertainty on M_{BH} (see the text for further detail).

bright sources (Bouquin et al. 2018). The disk galaxies have uncertainties due to our dust correction. There is also a need to account for uncertainties on the 3.6 μm B/T and D/T (Salo et al. 2015) used to covert the asymptotic magnitudes into disk, bulge, and total magnitudes. For the UV magnitudes, an additional source of systematic uncertainty is the derivation of the B/T and D/T (Section 3.1). The total uncertainties associated with the UV and 3.6 μm magnitudes were calculated by adding the individual contributions to the error budget in quadrature. Appendix C lists total magnitudes and associated errors for our sample galaxies. The quoted magnitudes are in the AB system unless noted otherwise.

3.3. Regression Analysis

Inherent differences in the employed statistical methods may systematically affect the derived black hole scaling relations, and it is therefore vital to explore this issue. We performed linear regression fits to $M_{BH} - \mathcal{C}(=m_{UV}^{corr} - m_{3.6}^{corr})$, $M_{BH} - L_{3.6}$, and $M_{BH} - \sigma$ data using three traditional regression

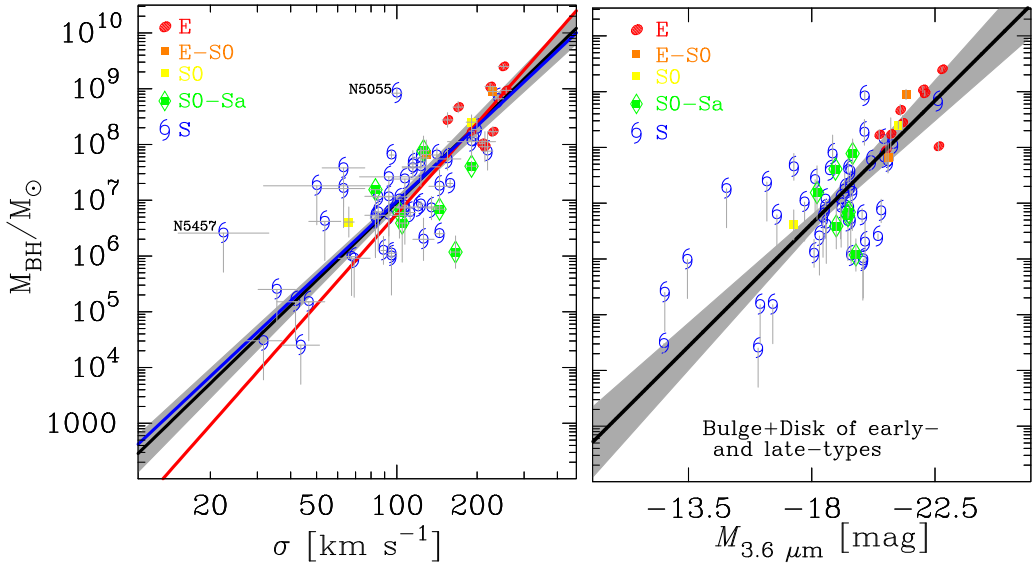

Figure 3. Similar to Figure 2, but we show here the correlations between $M_{\rm BH}$ and (left panel) velocity dispersion (σ, van den Bosch 2016, his Table 2) and (right panel) total 3.6 μm absolute magnitude of our sample galaxies ($M_{3.6\mu m}$). $M_{3.6\mu m}$ is computed using the total 3.6 μm apparent magnitudes ($m_{3.6\mu m}$, Table C1) and distances for the galaxies from van den Bosch (2016, his Table 2). We did not fit separate linear regressions to our early- and late-type ($M_{\rm BH}$, $M_{3.6\mu m}$) data or to the core-Sérsic and Sérsic ($M_{\rm BH}$, $M_{3.6\mu m}$) data, see the text for more details.

techniques: the bivariate correlated errors and intrinsic scatter (BCES) code (Akritas & Bershady 1996), the Bayesian linear regression routine (LINMIX_ERR, Kelly 2007), and the ordinary least-squares (OLS) code (Feigelson & Babu 1992); see Figures 2 and 3. The BCES routine (Akritas & Bershady 1996) was implemented in our work using the python module by Nemmen et al. (2012). While the BCES and LINMIX_ERR methods take into account the intrinsic scatter and uncertainties in $M_{\rm BH}$ and the host galaxy properties, only the latter can process "censored" data, e.g., black hole mass upper limits. The OLS routine does not account for errors; we use this method to provide $M_{\rm BH} - \mathcal{C}$ relations independent of measurement errors.

In an effort to assess the robustness of the above linear regression fits, we additionally performed symmetric MCMC Bayesian linear regression fits for our ($M_{\rm BH}$, $\mathcal{C}$) data, which account for black hole mass upper limits (see Appendix A).

Fitting the $Y = \beta X + \alpha$ line with the BCES, LINMIX_ERR, and OLS codes, we present the results from the $Y|X$, $X|Y$, and bisector regression analyses (Tables 1–3). We also present the results from our symmetric MCMC Bayesian regressions (Table 1). The $Y|X$ and $X|Y$ regressions minimize the residuals around the fitted regression lines in the Y and X directions, respectively. The symmetrical bisector line bisects the $Y|X$ and $X|Y$ lines. The LINMIX_ERR code does not return bisector regressions, thus we computed the the slope of the line that bisects the $Y|X$ and $X|Y$ lines. Throughout this work, we focus on the relations from the symmetrical BCES bisector regressions (Figures 2 and 3).

3.4. The $M_{BH} - \mathcal{C}_{\rm UV,tot}$ Relations

In this section, we investigate the correlations between $M_{\rm BH}$ and total (i.e., bulge+disk) colors $\mathcal{C}_{\rm FUV,tot}$ ($=m^{\rm corr}_{\rm FUV,total} - m^{\rm corr}_{\rm 3.6,total}$) and $\mathcal{C}_{\rm NUV,tot}$ ($=m^{\rm corr}_{\rm NUV,total} - m^{\rm corr}_{\rm 3.6,total}$) for our sample of 67 galaxies that is comprised of 20 early-type galaxies and 47 late-type galaxies (Appendix C). In Figure 2 we plot these correlations (Table C1) with data points that are color-coded based on morphological type. The regression analyses reveal that early- and late-type galaxies define two distinct red and blue sequences with markedly different slopes in the $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ diagrams, regardless of the applied regression methods (Table 1). We find that the slopes for early- and late-type galaxies are different at the $\sim 2\sigma$ level (Appendix A), and the significance levels for rejecting the null hypothesis that these two morphological types have the same slope are 1.7%–6.7%. We note in passing that this trend of different slopes for early- and late-type galaxies holds for the correlations between $M_{\rm BH}$ and the bulge colors of the two Hubble types ($\mathcal{C}_{\rm UV,bulge}$, Dullo et al. 2020). For both early- and late-type galaxies, the $\mathcal{C}_{\rm FUV,tot}$ and $M_{\rm BH}$ data correlate with a Pearson correlation coefficient $r \sim 0.60$ (Table 1). The $\mathcal{C}_{\rm NUV,tot}$ and $M_{\rm BH}$ data have Pearson correlation coefficients $r \sim 0.70$ and 0.65 for early- and late-type galaxies, respectively. In the $M_{\rm BH} - \mathcal{C}$ regression analyses, we have excluded one early-type galaxy (NGC 2685) and 2 late-type galaxies (NGC 3310 and NGC 4826) that are offset from the relations by more than three times the intrinsic scatter (Figure 2). The early-type galaxy NGC 5018 was also excluded from the $M_{\rm BH} - \mathcal{C}_{\rm FUV}$ relations, resulting in 18 early-type galaxies. These four outliers have peculiar characteristics that are discussed in Appendix B.1.

Symmetrical BCES bisector fits to the ($M_{\rm BH}$, $\mathcal{C}_{\rm FUV,tot}$) and ($M_{\rm BH}$, $\mathcal{C}_{\rm NUV,tot}$) data yield relations for early-type galaxies with slopes of 1.75 ± 0.41 and 1.95 ± 0.28, respectively (Table 1), such that $M_{\rm BH} \propto (L_{\rm FUV,tot}/L_{\rm 3.6,tot})^{-4.38\pm1.03}$ and $M_{\rm BH} \propto (L_{\rm NUV,tot}/L_{\rm 3.6,tot})^{-4.88\pm0.70}$. This is to be compared with the derived BCES bisector $M_{\rm BH} - \mathcal{C}_{\rm FUV,tot}$ and $M_{\rm BH} - \mathcal{C}_{\rm NUV,tot}$ relations for the late-type galaxies having shallower slopes of 1.03 ± 0.13 and 1.38 ± 0.23, such that $M_{\rm BH} \propto (L_{\rm FUV,tot}/L_{\rm 3.6,tot})^{-2.58\pm0.33}$ and $M_{\rm BH} \propto (L_{\rm NUV,tot}/L_{\rm 3.6,tot})^{-3.45\pm0.58}$.

With the assumption that the UV-to-3.6 μm luminosity ratio ($L_{\rm UV}/L_{\rm 3.6}$) is a proxy for the sSFR, (e.g., Bouquin et al. 2018), this implies that the growth of black holes in late-type galaxies has a steeper dependence on sSFR (i.e., $M_{\rm BH} \propto {\rm sSFR}^{-2.58}_{\rm FUV}$) than early-type galaxies ($M_{\rm BH} \propto {\rm sSFR}^{-4.38}_{\rm FUV}$). That is, at a given value of the sSFR, late-type galaxies tend to have more massive black holes than early-type galaxies. The caveat of using FUV magnitudes as a proxy for the current SFR is that a significant

Table 2
$M_{\rm BH} - \sigma$ Relation

Regression method	$Y\|X$		$X\|Y$		Bisector		r	Δ (dex)	ϵ (dex)	N
	α	β	α	β	α	β				
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
$Y = \beta X + \alpha$, $X = \log \sigma - 2.2$, $Y = \log M_{\rm BH}$ (early-type, Figure 3, left)										
BCES	7.84 ± 0.14	4.33 ± 0.80	7.81 ± 0.23	7.10 ± 1.58	**7.84 ± 0.16**	**5.42 ± 0.90**	0.72	0.67	0.70 ± 0.16	20
$Y = \beta X + \alpha$, $X = \log \sigma - 2.0$, $Y = \log M_{\rm BH}$ (late-type, Figure 3, left)										
	$Y\|X$		$X\|Y$		Bisector		r	Δ (dex)	ϵ (dex)	N
	α	β	α	β	α	β				
BCES	6.98 ± 0.10	4.00 ± 0.53	7.00 ± 0.11	5.00 ± 0.99	**6.98 ± 0.10**	**4.49 ± 0.48**	0.75	0.70	0.63 ± 0.10	45
$Y = \beta X + \alpha$, $X = \log \sigma - 2.0$, $Y = \log M_{\rm BH}$ (all galaxies, Figure 3, left)										
	$Y\|X$		$X\|Y$		Bisector		r	Δ (dex)	ϵ (dex)	N
	α	β	α	β	α	β				
BCES	6.99 ± 0.09	4.23 ± 0.40	6.92 ± 0.10	5.39 ± 0.72	**6.96 ± 0.09**	**4.65 ± 0.35**	0.78	0.68	0.62 ± 0.07	65

Note. Similar to Table 1, but here showing linear regression analyses of the correlation between SMBH mass ($M_{\rm BH}$) and velocity dispersion (σ).

fraction of the FUV light in ∼5% of massive early-type galaxies may come from extreme horizontal branch stars and not from young upper main-sequence stars. This phenomenon is called the "UV upturn" (e.g., Code & Welch 1979; O'Connell 1999; Yi et al. 2011). This is likely due to the rarity of very massive early-type galaxies in our sample. Nonetheless, we found that none of our early-type galaxies are UV upturns when we used the criteria given by Yi et al. (2011, their Table 1). Because the FUV band is more sensitive to the galaxy SFR than the NUV band, the $M_{\rm BH} - \mathcal{C}_{\rm FUV,tot}$ relations are systematically shallower than the corresponding $M_{\rm BH} - \mathcal{C}_{\rm NUV,tot}$ relations, although they are consistent with overlapping 1σ uncertainties (Table 1).

The root-mean-square (rms) scatter (Δ) around the fitted BCES bisector relations in the log $M_{\rm BH}$ direction is $\Delta_{\rm FUV,early} \sim 0.86$ dex, $\Delta_{\rm FUV,late} \sim 0.87$ dex, $\Delta_{\rm NUV,early} \sim 0.72$ dex, and $\Delta_{\rm NUV,late} \sim 0.86$ dex. We report intrinsic scatters (ϵ) for our $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ relations as derived by LINMIX_ERR to be $\epsilon_{\rm FUV,early} \sim 0.69 \pm 0.20$, $\epsilon_{\rm FUV,late} \sim 0.69 \pm 0.22$, $\epsilon_{\rm NUV,early} \sim 0.66 \pm 018$, and $\epsilon_{\rm NUV,late} \sim 0.66 \pm 0.18$.

As noted previously, the LINMIX_ERR code and MCMC Bayesian analysis—which account for the 24 galaxies (22 late-type galaxies and 2 early-type galaxies) with $M_{\rm BH}$ upper limits —yield $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ relations consistent with the BCES regression analyses (Table 1). Nonetheless, we checked for a potential bias for the late-type galaxies due to the inclusion of $M_{\rm BH}$ upper limits by rerunning the BCES bisector regression analysis on the 23 (=45–22) late-type galaxies with more securely measured $M_{\rm BH}$. We find that the slopes, intercepts, and Δ of the $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ relations are only weakly affected by the exclusion of $M_{\rm BH}$ upper limits.[6] Including the upper limits in the black hole scaling relations is useful (Gültekin et al. 2009) because they also follow the $M - \sigma$ relation that is traced by galaxies with more securely measured $M_{\rm BH}$ (Figure 3).

We can compare our work to that of Terrazas et al. (2017, their Figures 1 and 2), who used SFRs determined based on IRAS far-infrared imaging and reported an inverse correlation between sSFR and SMBH mass for 91 galaxies with measured black hole masses. Although they did not separate the galaxies into late- and early-types, their full sample seems to follow a single $M_{\rm BH}$ − sSFR relation with no break. This contradicts our results. To explain this discrepancy, we split the galaxies in Terrazas et al. (2017) by morphology and find that their late-type galaxies (which constitute one-third of the full sample) reside at the low-mass end of their $M_{\rm BH}$−sSFR relation and span very small ranges in SMBH mass ($4 \times 10^6 \lesssim M_{\rm BH}/M_\odot \lesssim 10^8$) and in sSFR ($10^{-11} \lesssim {\rm sSFR/yr^{-1}} \lesssim 8 \times 10^{-9}$). This is inadequate to establish the blue $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ sequence (Figure 2). Furthermore, we note that the far-IR (FIR) flux may underestimate the actual SFR for low-mass late-type galaxies because most of the UV photons of these galaxies are unobscured by dust and are thus not reprocessed to FIR wavelengths (Catalán-Torrecilla et al. 2015). We also find that the Terrazas et al. (2017) early-type $M_{\rm BH}$ − sSFR relation has more scatter than ours. We suspect that this may be due to a variable contamination of the IRAS FIR emissions in massive galaxies by the AGN[7] that heats the surrounding dust. While the dusty AGN in some massive galaxies might led to an increase in the SFR values based on FIR luminosities (e.g., Catalán-Torrecilla et al. 2015; Toba et al. 2017), star formation activities are likely the dominant contributor to the SFR values reported by Terrazas et al. (2017) because AGN contamination is mainly responsible for the larger scatter observed in their $M_{\rm BH}$ − sSFR relation.

3.5. $M_{BH} - \sigma$ and M_{BH} − L Relations

In Figure 2 we have shown for the first time, to our knowledge, a correlation between SMBH mass and total (i.e., bulge+disk) colors for early- and late-type galaxies. For this correlation to be evident, it is important that the color is determined using a wide wavelength baseline. In this section, we present correlations between $M_{\rm BH}$ and velocity dispersion (σ) and 3.6 μm total luminosity ($L_{3.6,\rm tot}$) for the same galaxy sample to allow a direct statistical comparison with the

[6] The strength of the $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ correlations decreases when the SMBH upper limits are excluded. The Pearson correlation coefficient for the ($\mathcal{C}_{\rm FUV,tot}$, $M_{\rm BH}$) blue sequence has reduced from $r \sim 0.60$ to 0.34 due to the exclusion of $M_{\rm BH}$ upper limits, and for the ($\mathcal{C}_{\rm NUV,tot}$, $M_{\rm BH}$) blue sequence, there is a decrease in r from ∼0.65 to 0.36.

[7] While the IR emissions in massive galaxies have contributions from the dusty AGN, the NUV emissions arise from main-sequence turn-off stars and are less prone to contamination from the AGN (Boselli et al. 2005).

Table 3
$M_{\rm BH} - L_{3.6,\rm tot}$ Relation

	$Y = \beta X + \alpha$, $X = M_{3.6,\rm tot} + 18.5$, $Y = \log M_{\rm BH}$ (all galaxies, Figure 3, right)									
Regression method	$Y\|X$		$X\|Y$		Bisector		r	Δ (dex)	ϵ (dex)	N
	α	β	α	β	α	β				
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
BCES	6.96 ± 0.11	−0.37 ± 0.06	6.80 ± 0.17	−0.65 ± 0.12	**6.88 ± 0.12**	**−0.49 ± 0.06**	−0.66	0.84	0.69 ± 0.09	67

Note. Similar to Table 1, but here showing linear regression analyses of the correlation between the SMBH masses ($M_{\rm BH}$) and the 3.6 μm total absolute magnitude of the galaxies ($M_{3.6,\rm tot}$), see the text for details.

$M_{\rm BH} - \mathcal{C}$ relations (Figure 3 and Table 2). Assuming a 10% uncertainty[8] on σ, the BCES bisector regression yields $M_{\rm BH} - \sigma$ relations with slopes 5.42 ± 0.90, 4.49 ± 0.48, and 4.65 ± 0.35 for the early-type galaxies and late-type galaxies, and for the full ensemble. These relations are in good agreement with each other within their 1σ uncertainties. The late-type galaxies NGC 5055 and NGC 5457, which are the most deviant outliers in the $M_{\rm BH} - \sigma$ diagram, were excluded. The unification of early- and late-type galaxies in the $M_{\rm BH} - \sigma$ diagram has been reported before (e.g., Beifiori et al. 2012; Graham & Scott 2013; van den Bosch 2016; Dullo et al. 2020, submitted). The $M_{\rm BH} - \sigma$ relations (Table 2) are consistent with those from Gültekin et al. (2009), Beifiori et al. (2012), McConnell & Ma (2013), Kormendy & Ho (2013), and Graham & Scott (2013).[9]

To determine the $M_{\rm BH} - L_{3.6,\rm tot}$ relation, we converted the inclination and dust-corrected 3.6 μm total apparent magnitudes into absolute magnitudes ($M_{3.6,\rm tot}$, Appendix C) using distances given in van den Bosch (2016). The BCES bisector $M_{\rm BH} - L_{3.6,\rm tot}$ relation for the full sample of 67 galaxies was performed without accounting for the error on $M_{3.6,\rm tot}$, yielding $M_{\rm BH} \propto L_{3.6,\rm tot}^{1.23\pm0.15}$ (Figure 3 and Table 3). Graham & Scott (2013) reported two distinct $M_{\rm BH} - L$ relations for the bulges of Sérsic galaxies ($M_{\rm BH} \propto L_{\rm K_s,bulge}^{2.73}$) and core-Sérsic galaxies ($M_{\rm BH} \propto L_{\rm K_s,bulge}^{1.10}$). Because only 7 galaxies were identified as core-Sérsic galaxies (see Figure 4), we refrained from separating the galaxies into Sérsic and core-Sérsic galaxies in the $M_{\rm BH} - L_{3.6,\rm tot}$ diagram.

4. Discussion

4.1. Comparison Between the $M_{\rm BH} - \mathcal{C}$, $M_{BH} - \sigma$ and $M_{BH} - L$ Relations

Because a different number of galaxies was used to define the $M_{\rm BH} - \mathcal{C}$, $M_{\rm BH} - \sigma$ and $M_{\rm BH} - L$ relations, a direct comparison of the strength and scatter of the relations is difficult. Nonetheless, the correlation between the color $\mathcal{C}_{\rm UV,tot}$ and black hole mass $M_{\rm BH}$ ($r \sim 0.60$–0.70, see Table 1) is slightly weaker than that between the stellar velocity dispersion σ and $M_{\rm BH}$ ($r \sim 0.72$–0.78, Table 2). These two relations have comparable intrinsic scatter (see Tables 1 and 2). In terms of scatter in the log $M_{\rm BH}$ direction, the $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ relations

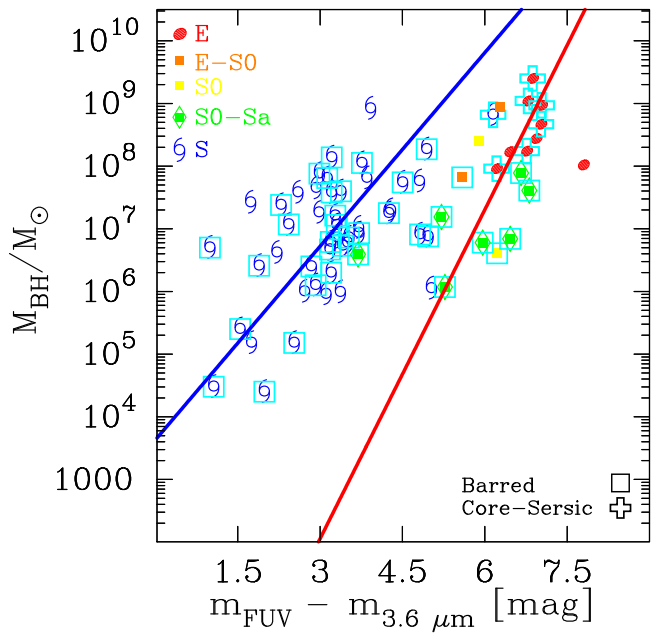

Figure 4. Similar to Figure 2(a), but here we also show the host galaxy properties. Barred galaxies are enclosed in boxes. Seven core-Sérsic galaxies (6 Es + 1 S) with partially depleted cores published in the literature are enclosed in crosses (see Section 4.2).

typically have 5%–27% more scatter (i.e., $\Delta \sim 0.72$–0.87 dex) than the $M_{\rm BH} - \sigma$ relations ($\Delta \sim 0.68$–0.70 dex). The $M_{\rm BH} - \sigma$ relation appears to be the most fundamental SMBH scaling relation. However, the $M_{\rm BH} - \sigma$ relations for late- and early-type galaxies are not notably offset from each other. This contrasts with the formation models of galaxies, which predict that the SMBH growth in the two Hubble types is completely different (see Section 4.5). Furthermore, Dullo et al. (2020, submitted) showed that the $M_{\rm BH} - \sigma$ relation tends to underpredict the actual black hole masses for the most massive galaxies with $M_* \gtrsim 10^{12} M_\odot$. In contrast, the $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ relations are in accordance with galaxy formation models (Section 4.5).

In the comparison between the $M_{\rm BH} - \mathcal{C}$ and $M_{\rm BH} - L$ relations, these two relations display a similar level of strength and vertical scatter (Table 3, $r \sim 0.60$–0.70 and $\Delta \sim 0.72$–0.87 dex for $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ and $r \sim -0.66$, and $\Delta \sim 0.84$ dex for $M_{\rm BH} - L_{3.6,\rm tot}$). In passing we note that the existence of the $M_{\rm BH} - L$ relation coupled with the red sequence and blue cloud traced by early- and late-type galaxies in the color–magnitude diagram does not necessitate a correlation between the black hole mass and galaxy color.

For comparison, our $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ relation for the 47 late-type galaxies is stronger ($r \sim 0.60$–0.65, and $\Delta \sim 0.87$ dex)

[8] After comparing the HyperLeda individual velocity dispersion measurements and mean homogenized values for 100 sample galaxies, we adopt a conservative upper limit on the uncertainty of 10% on σ.

[9] Our slopes for the full sample of 66 galaxies are slightly (i.e., $\sim 1.4\sigma \approx 15\%$) shallower than the slope reported by van den Bosch (2016, slope $\sim 5.35 \pm 0.23$). This is because the van den Bosch (2016, his Figure 1) sample contains extremely bright galaxies with $\sigma \gtrsim 270$, which tend to steepen the $M_{\rm BH} - \sigma$ relation (see Figure 1).

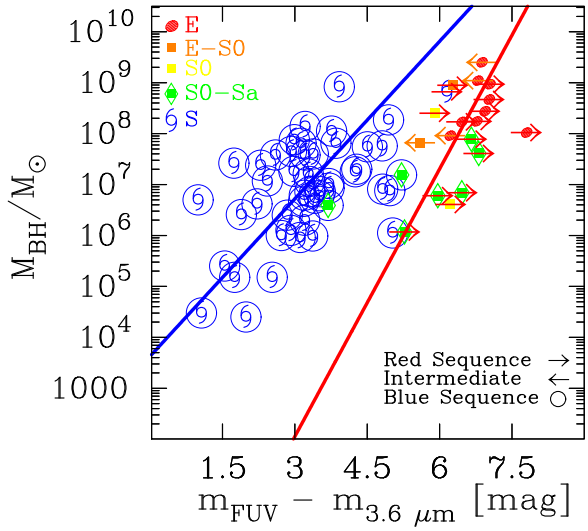

Figure 5. Similar to Figure 2, but here comparing the $M_{BH} - \mathcal{C}$ red and blue sequences and the color–color red/intermediate and blue sequences. The color–color relation red- and intermediate-sequence galaxies (Bouquin et al. 2018, their Section 4.4) are marked by rightward- and leftward-pointing arrows, respectively. Blue-sequence galaxies are enclosed in circles.

than the $M_{BH} - M_{*,tot}$ relation based on 3.6 μm data by Davis et al. (2018) for their sample of 40 late-type galaxies ($r \sim 0.47$ and $\Delta \sim 0.79$ dex).

4.2. Core-Sérsic versus Sérsic

As noted in the Introduction, the SMBH scaling relations may differ depending on the galaxy core structure (i.e., core-Sérsic versus Sérsic type). We identify seven core-Sérsic galaxies (6 Es + 1 S) in our sample with partially depleted cores published in the literature: NGC 1052 (Lauer et al. 2007), NGC 3608, NGC 4278, and NGC 4472 (Dullo & Graham 2012, 2014), and NGC 4374, NGC 4594, and NGC 5846 (Graham & Scott 2013). They are among the reddest ($\mathcal{C}_{FUV,tot} \gtrsim 6$) galaxies in our sample with massive SMBHs ($M_{BH} \gtrsim 10^8 M_\odot$), Figure 4. While structural analyses of high-resolution Hubble Space Telescope images are needed to identify a partially depleted core (or lack thereof) in the remaining sample galaxies (e.g., Dullo & Graham 2013, 2014; Dullo et al. 2016, 2017, 2018, the majority (~80%) of our spiral galaxies have $\sigma \lesssim 140$ km s^{-1}, and they are likely Sérsic galaxies with no partially depleted cores (e.g., Dullo & Graham 2012, 2013, 2014; Dullo et al. 2017). For early-type galaxies, we did not find bends or offsets from the $M_{BH} - \mathcal{C}_{UV,tot}$ relations because of core-Sérsic or Sérsic galaxies (Figure 4).

4.3. Red, Intermediate, and Blue Sequences

To locate our galaxies in color–color diagrams, we used the classification by Bouquin et al. (2018, their Section 4.4). These authors compared the FUV−NUV and NUV–[3.6] colors to separate their galaxies into red, intermediate, and blue sequences. Figure 5 shows excellent coincidence between the red and blue $M_{BH} - \mathcal{C}$ early- and late-type morphological sequences (Sections 3.4) and the canonical color–color relation (red/intermediate) and blue sequences, respectively, with only three exceptions (NGC 2685, NGC 4245, and NGC 4594). The case of NGC 2685 was discussed in Appendix B.1.4. NGC 4245 is a barred S0-Sa galaxy with a prominent ring

Table 4
SMBH Masses

Galaxy (1)	Type (2)	$\mathcal{C}_{UV,tot}$ (AB mag) (FUV/NUV) (3)	log ($M_{BH}/M_\odot$) (directly measured) (4)	log ($M_{BH}/M_\odot$) (predicted) (FUV/NUV) (5)
NGC 0205	E5 pec	4.61/2.67	$3.83^{+1.18}_{-1.83}$	4.85/3.58
NGC 0524	SA0	7.30/5.48	$8.94^{+0.05}_{-0.05}$	9.68/9.02
NGC 0821	E6	7.02/4.69	$8.22^{+0.21}_{-0.21}$	9.13/7.64
NGC 1023	SB0	6.64/5.01	$7.62^{+0.05}_{-0.05}$	8.39/8.20
NGC 4395	SAm	1.49/1.37	$5.54^{+0.54}_{-0.54}$	5.17/5.29
NGC 4459	SA0	6.56/5.03	$7.84^{+0.09}_{-0.09}$	8.24/8.23
NGC 4473	E5	6.80/4.93	$7.95^{+0.24}_{-0.24}$	8.71/8.06
NGC 4552	E	6.24/5.33	$8.70^{+0.05}_{-0.05}$	7.61/8.76
NGC 4564	E6	6.34/4.77	$7.95^{+0.12}_{-0.12}$	7.91/7.78
NGC 4621	E5	7.18/5.78	$8.60^{+0.09}_{-0.09}$	9.41/9.55
NGC 5845	E	6.38/5.30	$8.69^{+0.16}_{-0.16}$	7.89/8.71

Note. Column (1) lists the galaxy name. Column (2) lists the morphological type from the NED database. Column (3) lists the total FUV–[3.6] and NUV–[3.6] colors ($\mathcal{C}_{FUV,tot}$ and $\mathcal{C}_{NUV,tot}$). Columns (4) list the directly measured SMBH masses. Column (5) lists black hole masses predicted using the $M_{BH} - \mathcal{C}_{FUV,tot}$ and $M_{BH} - \mathcal{C}_{NUV,tot}$ relations (Table 1) and the appropriate colors given in Column (3). We assign a typical uncertainty of 0.85 dex to log (M_{BH}) for these predicted black hole masses.

(Treuthardt et al. 2007). The spiral Sa NGC 4594 (also referred to as the Sombrero Galaxy) exhibits properties similar to massive early-type galaxies. Spitler et al. (2008) found that the number of blue globular clusters in NGC 4594 is comparable to massive early-type galaxies. Moreover, Jardel et al. (2011) noted that the galaxy dark matter density and core radius resemble those expected for early-type galaxies with massive bulges. It is the only red-sequence spiral in our sample (Bouquin et al. 2018), which is also unique in being the only core-Sérsic late-type galaxy in the sample. Interestingly, Figure 5 reveals that all the intermediate-sequence galaxies (Bouquin et al. 2018) reside to the left of the $M_{BH} - \mathcal{C}_{UV,tot}$ relation that is defined by early-type galaxies.

4.4. Predicting SMBH Masses Using the $M_{BH} - \mathcal{C}_{UV}$ Relations

It is of interest to assess the robustness of black hole masses that are estimated using the $M_{BH} - \mathcal{C}_{FUV,tot}$ and $M_{BH} - \mathcal{C}_{NUV,tot}$ relations found in this work. We do so using literature FUV, NUV, and 3.6 μm magnitudes for a selected sample of 11 galaxies with direct SMBH masses that are not in our sample (see Table 4). These 11 galaxies were not included in the main sample because we endeavor to establish the $M_{BH} - \mathcal{C}_{UV,tot}$ relations using homogeneously determined UV and 3.6 μm magnitudes. In this way, the observed trends in the $M_{BH} - \mathcal{C}_{UV,tot}$ diagrams (Figure 2) cannot be attributed to differences in methods and/or data sources. We use the black hole mass measurement of NGC 205 by Nguyen et al. (2019), and for the remaining 10 galaxies, the SMBH masses are taken from van den Bosch (2016). Of the 11 galaxies, 9 are in common between Jeong et al. (2009) and Savorgnan & Graham (2016). Jeong et al. (2009, their Table 1) published total apparent FUV and NUV magnitudes derived from growth curves for the galaxies, while Savorgnan & Graham (2016,

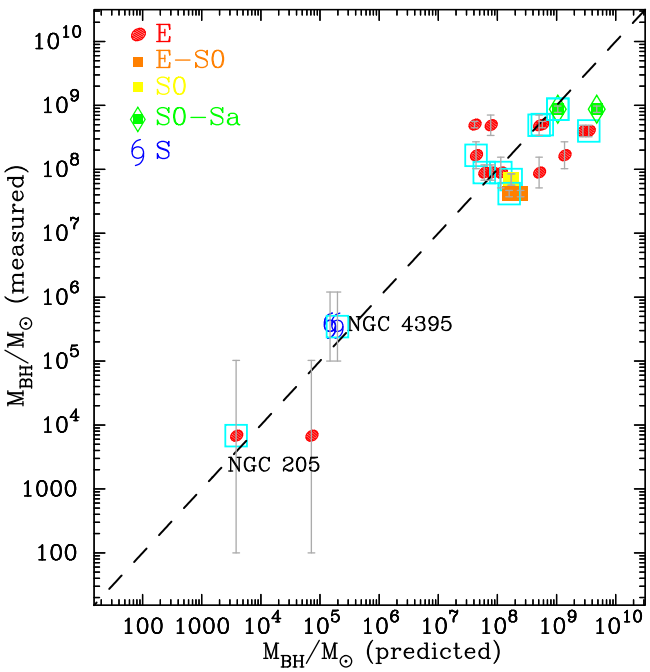

Figure 6. Comparison between SMBH masses predicted using our $M_{\rm BH} - \mathcal{C}_{\rm FUV,tot}$ and $M_{\rm BH} - \mathcal{C}_{\rm NUV,tot}$ relations (Table 1) and those determined dynamically (van den Bosch 2016; Nguyen et al. 2019) for a selected sample of 11 galaxies that are not in our sample (see Table C1). SMBH masses predicted based on the $M_{\rm BH} - \mathcal{C}_{\rm NUV,tot}$ relations are enclosed in boxes.

their Table 2) presented their 3.6 μm galaxy apparent magnitudes[10] that were computed using their best-fitting structural parameters. We also included the low-mass elliptical galaxy NGC 205 and the Seyfert SAm bulgeless galaxy NGC 4395 (Filippenko & Ho 2003; Peterson et al. 2005). NGC 205 potentially harbors the lowest central black hole mass measured for any galaxy to date (Nguyen et al. 2019), and NGC 4395 is known to be an outlier from the $M_{\rm BH} - \sigma$ diagrams (e.g., Davis et al. 2017). For NGC 205, we use the UV and 3.6 μm magnitudes from Gil de Paz et al. (2007) and Marleau et al. (2006), respectively. For NGC 4395, the total UV and 3.6 μm magnitudes are from Dale et al. (2009) and Lee et al. (2011); a caveat here is that these magnitudes are not corrected for internal dust attenuation.

Before applying our $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ relations to estimate $M_{\rm BH}$, we homogenize the FUV, NUV, and 3.6 μm data from the literature by comparing our magnitudes with those from Jeong et al. (2009) and Savorgnan & Graham (2016) for galaxies in common with them. We find that compared to our magnitudes, the Jeong et al. (2009) total FUV and NUV magnitudes are fainter by typically 0.52 mag, while the galaxy magnitudes from Savorgnan & Graham (2016) are brighter by typically 0.18 mag. Having applied these corrections (i.e., $m_{\rm UV} = m_{\rm UV,Jeo} - 0.52$ and $m_{3.6} = m_{3.6,\rm Sav} + 0.18$) for the 9 early-type galaxies, we computed the $\mathcal{C}_{\rm FUV,tot}$ and $\mathcal{C}_{\rm FUV,tot}$ colors listed in Table 4.

Figure 6 reveals good agreement between the directly measured $M_{\rm BH}$ and predicted $M_{\rm BH}$ determined using $M_{\rm BH} - \mathcal{C}_{\rm FUV,tot}$ and $M_{\rm BH} - \mathcal{C}_{\rm NUV,tot}$ relations for the 10 galaxies in Table 4. On average, $|\log(M_{\rm BH,predicted}/M_{\rm BH,measured})| \sim 0.67$ dex $\pm$ 0.29 dex(FUV) and $\sim$0.32 dex $\pm$ 0.29 dex (NUV). In Figure 6 the direct black hole mass appears to correlate better with that predicted using the NUV color than using the FUV color, and for the massive early-type galaxies, this may be due to contributions to the FUV flux from the extreme horizontal branch stars (see Section 3.4). The approach of using homogenized galaxy colors obtained through different methods may introduce some systematic errors in the determination of $M_{\rm BH}$. We caution that when the $M_{\rm BH} - \mathcal{C}$ relations are used to predict black hole masses, one should use FUV, NUV, and 3.6 μm magnitudes that are obtained in a homogeneous way. Furthermore, the $M_{\rm BH} - \mathcal{C}$ relations should not be used to predict black hole masses in galaxies that are highly inclined (e.g., edge-on) and obscured with dust.

As noted in the introduction, a clear benefit of the $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ relation is its applicability to early- and late-type galaxies, including those with low central velocity dispersions ($\sigma \lesssim 100$ km s^{-1}) and with small or no bulges. Moreover, photometry has the advantage of being cheaper than spectroscopy. Using our relations (Table 1) together with galaxy colors derived from the Bouquin et al. (2018, their Table 1) asymptotic FUV, NUV, and 3.6 μm magnitudes, we tentatively predict black hole masses in a sample of 1382 GALEX/S^{4}G galaxies (Table D1) with no measured black hole masses, see Appendix D. We show that late-type galaxies with $\mathcal{C}_{\rm FUV,tot} \lesssim 1.33$ AB mag or $\mathcal{C}_{\rm NUV,tot} \lesssim 1.28$ AB mag may harbor IMBHs ($M_{\rm BH} \sim 100$–$10^5 M_\odot$). Similarly, early-type galaxies with $\mathcal{C}_{\rm FUV,tot} \lesssim 4.68$ AB mag or $\mathcal{C}_{\rm NUV,tot} \lesssim 3.4$ AB mag are potential IMBH hosts. While Sloan Digital Sky Survey (SDSS) velocity dispersion measurements are available for hundreds of thousands of galaxies and one can use them together with the $M_{\rm BH} - \sigma$ relation to estimate black hole masses, as cautioned by the SDSS Data Release[11] 12 (Alam et al. 2015), velocity dispersion values lower than 100 km s^{-1} reported by SDSS are below the resolution limit of the SDSS spectrograph and are regarded as unreliable. Note that galaxies with $\sigma \lesssim 100$ km s^{-1} are expected to have low stellar masses ($M_* \lesssim 2 \times 10^{10} M_\odot$), and such galaxies make up a significant fraction of the SDSS galaxy sample (Chang et al. 2015, see their Figure 9). In addition, the SDSS spectra measure the light within a fixed aperture of radius 1.″5, thus the SDSS velocity dispersion values of more distant galaxies can be systematically lower than those of similar nearby galaxies.

4.5. The $M_{\rm BH} - \mathcal{C}_{\rm UV}$ Correlation as Evidence for the Coevolution of SMBHs and Galaxies

By morphologically splitting the sample galaxies, we have demonstrated that late-type hosts do not correlate with SMBHs in the same manner as early-type hosts (Table 1). This can be reconciled very well with the prediction that early- and late-type galaxies have fundamentally different formation histories (e.g., White & Rees 1978; Khochfar & Burkert 2001; Steinmetz & Navarro 2002; Kauffmann et al. 2003b; Kormendy & Kennicutt 2004; Schawinski et al. 2010, 2014; Dullo & Graham 2014; Tonini et al. 2016; Davis et al. 2018; Dullo et al. 2019). For example, the red and blue $M_{\rm BH} - \mathcal{C}_{\rm FUV,tot}$ relations are such that $M_{\rm BH,early\text{-}type} \propto (L_{\rm FUV,tot}/L_{3.6,\rm tot})^{-4.38}$ and $M_{\rm BH,late\text{-}type} \propto (L_{\rm FUV,tot}/L_{3.6,\rm tot})^{-2.58}$. Given that $L_{\rm UV}/L_{3.6}$ is a good proxy for the sSFR, (Bouquin et al. 2018. their Appendix B), this therefore implies that both early- and late-type galaxies exhibit log-linear inverse correlations between

[10] We have converted the 3.6 μm VEGA magnitudes from Savorgnan & Graham (2016) into AB magnitudes.

[11] https://www.sdss.org/dr12/algorithms/redshifts/

$M_{\rm BH}$ and sSFR, the latter having a steeper dependence on sSFR than the former (see Table 1 and Figure 2).

A correlation between $M_{\rm BH}$ and sSFR is expected. Observations have shown that bright quasars and local Seyferts tend to reside in strong starburst galaxies or in galaxies with ongoing star formation (e.g., Sanders et al. 1988; Kauffmann et al. 2003a; Alexander et al. 2005; Lutz et al. 2008; Netzer 2009; Wild et al. 2010; Rosario et al. 2012; Barrows et al. 2017; Yang et al. 2017). Other findings lending further support to the link between star formation and SMBH growth are the correlation between black hole accretion rate and host galaxy SFR (e.g., Heckman et al. 2004; Merloni et al. 2004; Hopkins & Quataert 2010; Chen et al. 2013; Madau & Dickinson 2014; Sijacki et al. 2015; Yang et al. 2017), the inverse correlation between sSFR and specific SMBH mass (Terrazas et al. 2017), and the trend between SMBH mass and host galaxy star formation histories over cosmic time (e.g., Martín-Navarro et al. 2018; van Son et al. 2019).

Within the self-regulated SMBH growth model, the correlation between SMBH masses and the host galaxy properties (e.g., stellar luminosity, McLure & Dunlop 2002; Marconi & Hunt 2003) is interpreted as reflecting a link between the growth of SMBHs and star formation events in the host (Silk & Rees 1998; Fabian 1999; King 2003; Di Matteo et al. 2005; Murray et al. 2005; Springel et al. 2005; Croton et al. 2006; Hopkins et al. 2006; Schawinski et al. 2006; Cattaneo et al. 2009; Weinberger et al. 2017). In this scenario, the same cold gas reservoir that fuels the AGN/quasar feeds starburst events. The energy or momentum released by the AGN/quasar can heat the interstellar medium and cause the expulsion of gas from the host galaxy, shutting off star formation and halting accretion onto the SMBH. However, whether AGN accretion and star formation are precisely coincidental is unclear (e.g., Ho 2005).

We (see also Dullo et al. 2020, submitted) argue that the significantly different $M_{\rm BH} - \mathcal{C}_{\rm FUV}$ relations for early- and late-type galaxies (i.e., the $M_{\rm BH} - \mathcal{C}_{\rm FUV}$ red and blue sequences) suggest that the two Hubble types follow two distinct channels of SMBH growth: the former is driven by a major merger, and the latter involves (major merger) free processes (see Hopkins & Hernquist 2009). This broadly agrees with Schawinski et al. (2010, 2014).

The standard cosmological formation paradigm is that SMBHs in early-type galaxies are built up during the period of rapid galaxy growth at high redshift ($z \sim 2$–5) when the major gas-rich mergers of disk galaxies (e.g., Toomre & Toomre 1972; White & Rees 1978) drive gas infall into the nuclear regions of the newly formed merger remnant, leading to starburst events and AGN accretion processes (e.g., Barnes & Hernquist 1991, 1996; Naab et al. 2006a; Hopkins & Quataert 2010; Knapen et al. 2015). Accretion onto more massive SMBHs triggers stronger AGN feedback, which is efficient at quenching star formation rapidly. The position of an early-type galaxy on the $M_{\rm BH} - \mathcal{C}_{\rm UV}$ red sequence is therefore dictated by the complex interplay between the details of its SMBH growth, efficiency of AGN feedback, regulated star formation histories, and major merger histories; this is not set by simple hierarchical merging (Peng 2007; Jahnke & Macciò 2011).

Massive early-type galaxies (i.e., total stellar mass $M_{*,k} \sim 8 \times 10^{10} M_\odot$–$10^{12} M_\odot$) with $M_{\rm BH} \gtrsim 10^8 M_\odot$ and $\mathcal{C}_{\rm FUV} \gtrsim 6.3$ mag, at the high-mass end of the $M_{\rm BH} - \mathcal{C}_{\rm UV}$ red sequence (Figure 2, Section 4.3), are consistent with the scenario where a (gas-rich) major merger at high redshift drives intense bursts of star formation, efficient SMBH growth, and ensuing quenching of star formation by strong AGN feedback on short timescales (e.g., Thomas et al. 2005, 2010; de La Rosa et al. 2011; McDermid et al. 2015; Segers et al. 2016). This is accompanied by a few (0.5–2) successive gas-poor (dry) major mergers since $z \sim 1.5$–2 (e.g., Bell et al. 2004, 2006; Khochfar & Silk 2009; Dullo & Graham 2012, 2013, 2014; Man et al. 2012; Rodriguez-Gomez et al. 2015), involving low level star formation[12] (i.e., "red but not strictly dead," see de La Rosa et al. 2011; Davis et al. 2019; Habouzit et al. 2019) detected by the GALEX FUV and NUV detectors (e.g., Gil de Paz et al. 2007; Bouquin et al. 2018). The bulk of these objects are core-Sérsic elliptical galaxies, a fraction of which may gradually grow stellar disk structures and transform into massive lenticular galaxies (Dullo & Graham 2013; Graham 2013; Dullo 2014; Graham et al. 2015; de la Rosa et al. 2016).

As for the less massive (Sérsic) early-type galaxies ($M_{*,k} \sim 10^{10} M_\odot - 2 \times 10^{11} M_\odot$) with lower SMBH masses ($10^6 M_\odot \lesssim M_{\rm BH} \lesssim 10^8 M_\odot$) and $\mathcal{C}_{\rm FUV} \lesssim 6.3$ mag (Figure 2, Section 4.3), they likely grow primarily via gas-rich (wet) major mergers and form their stellar populations over an extended period of time (Thomas et al. 2005, 2010; de La Rosa et al. 2011; McDermid et al. 2015). Our findings disfavor a scenario where Sérsic early-type galaxies with intermediate colors are late-type galaxies quenching star formation and moving away from the $M_{\rm BH} - \mathcal{C}_{\rm FUV}$ blue sequence. Collectively, core-Sérsic and Sérsic early-type galaxies define a red sequence in the $M_{\rm BH} - \mathcal{C}_{\rm FUV}$ diagram. Lacking the most luminous and massive BCGs with $M_{*,k} \gtrsim 10^{12} M_\odot$ in our sample, we note that our $M_{\rm BH} - \mathcal{C}_{\rm UV}$ relation is not constrained at the highest-mass end. Because such galaxies are generally expected to have negligible star formation (Dullo 2019), the $M_{\rm BH} - \mathcal{C}_{\rm UV}$ relations, the $M_{\rm BH} - \mathcal{C}_{\rm FUV}$ in particular, may not apply to them. Our interpretation of the assembly of the red sequence for early-type galaxies is in accordance with the "downsizing" scenario, where more massive galaxies form stars earlier and over a shorter timescale than less massive galaxies (e.g., Cowie et al. 1996; Brinchmann & Ellis 2000; Cattaneo et al. 2008; Pérez-González et al. 2008; Pannella et al. 2009).

The (major merger) free scenario—secular processes and minor mergers—may be naturally consistent with the observed late-type $M_{\rm BH} - \mathcal{C}_{\rm FUV}$ blue sequence (Figure 2, Section 4.3) and dynamically cold stellar disks of late-type galaxies. Recently, Martin et al. (2018a) reported that massive SMBHs in disk galaxies can grow primarily via secular processes with small contributions (i.e., only 35% of the SMBH mass) from mergers. We tentatively hypothesize that secular-driven processes involving nonaxisymmetric stellar structures, such as bars and spiral arms, can trigger a large inflow of gas from the large-scale disk into nuclear regions of late-type galaxies, slowly feeding SMBHs and fueling star formation (e.g., Kormendy 1993, 2013; Kormendy & Kennicutt 2004; Fisher & Drory 2008; Leitner 2012; Cisternas et al. 2013; Tonini et al. 2016; Dullo et al. 2019). Barred galaxies make up the bulk (~62%) of the late-type

[12] Massive early-type galaxies and some BCGs can acquire cold gas through the cooling of hot gas and/or via cannibalism of a gas-rich satellite, and they may undergo episodes of low level star formation at low redshift (Salomé & Combes 2003; O'Dea et al. 2008; Hopkins & Hernquist 2009; Struve et al. 2010; Young et al. 2011; Zubovas & King 2012; Russell et al. 2014, 2019; Smith & Edge 2017; Krajnović et al. 2020).

galaxies in our sample (Figure 4). In addition, gas-rich minor mergers have been suggested in the literature to trigger enhanced star formation and SMBH growth in late-type galaxies without destroying the disks (Simmons et al. 2013, 2017; Kaviraj 2014a, 2014b; Martin et al. 2018a, 2018b). The question remains, however, whether pure (major merger) free processes could be the main mechanism for the formation of late-type galaxies with massive bulges and high-velocity dispersion, as in the case of NGC 4594.

As noted above, we find that core-Sérsic[13] and Sérsic galaxies collectively define a single early- or late-type morphology sequence, in agreement with the conclusions by Sahu et al. (2019a), Dullo (2019), and Dullo et al. (2020, submitted). Moreover, Savorgnan et al. (2016) reported a blue spiral galaxy $M_{\rm BH} - M_{*,\rm bulge}$ sequence using their sample of 17 spiral galaxies (see also Davis et al. 2018, their Section 2.2).

We remark that bulgeless spirals and spiral galaxies with classical bulges or pseudo-bulges all follow the blue late-type $M_{\rm BH} - \mathcal{C}_{\rm UV}$ relations (Figure 2). Pseudo-bulges are typically hosted by late-type galaxies and a few early-type galaxies, while classical bulges are generally associated with early-type galaxies and massive late-type galaxies. A key point to note here is that the relatively high sSFR for pseudo-bulges coupled with the steeper dependence of SMBH masses on sSFRs for late-type galaxies likely explain why pseudo-bulges seem to obey a different $M_{\rm BH} - \sigma$ relation than classical bulges (e.g., Kormendy & Ho 2013). This also explains why bulgeless spirals and low-mass spirals are offset from the $M_{\rm BH} - L_{\rm Bulge}$, $M_{\rm BH} - M_{*,\rm Bulge}$, and $M_{\rm BH} - \sigma$ relations defined by the massive early- and late-type galaxies, as reported in the literature (e.g., Greene et al. 2008, 2010, 2016; Baldassare et al. 2017, Caglar et al. 2020).

5. Conclusions

Using a sample of 67 GALEX/S^{4}G galaxies with directly measured SMBH masses ($M_{\rm BH}$), comprised of 20 early-type galaxies and 47 late-type galaxies, we establish for the first time a correlation between ($M_{\rm BH}$) and the host galaxy total (i.e., bulge+disk) UV−[3.6] color ($\mathcal{C}_{\rm UV}$). More massive SMBHs are hosted by galaxies with redder colors. The GALEX FUV/NUV and S^{4}G Spitzer 3.6 μm asymptotic magnitudes of the sample galaxies determined in a homogeneous manner (Bouquin et al. 2018) along with their 3.6 μm multicomponent decomposition by Salo et al. (2015) were used to derive dust-corrected total (bulge+disk) magnitudes in FUV, NUV, and 3.6 μm bands. We provide these magnitudes in Table C1.

We fit our ($M_{\rm BH}$, $\mathcal{C}_{\rm UV,tot}$) data set using several statistical techniques, focusing on the symmetric BCES bisector regressions. Our key findings are as follows.

(1) Investigating the nature of the $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ relations, our results show that early-type galaxies define a red sequence in the $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ diagrams that is different from the late-type blue sequence. We found a strong tendency for the galaxies that lie on the red/blue $M_{\rm BH} - \mathcal{C}$ morphological sequences to also be on the (red plus intermediate)/blue sequences in the canonical color–color relation (see Section 4.3 and Figure 5).

(2) The $M_{\rm BH} - \mathcal{C}_{\rm FUV,tot}$ and $M_{\rm BH} - \mathcal{C}_{\rm NUV,tot}$ relations for early-type galaxies have slopes of 1.75 ± 0.41 and 1.95 ± 0.28, respectively, whereas for late-type galaxies, the slopes are substantially shallower, i.e., 1.03 ± 0.13 and 1.38 ± 0.23. The early- and late-type $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ relations have rms scatters (Δ) in the log $M_{\rm BH}$ direction of $\Delta_{\rm UV,early} \sim 0.72$–$0.86$ dex and $\Delta_{\rm UV,late} \sim 0.86$ dex and Pearson correlation coefficients (r) of $r_{\rm early} \sim 0.61$–0.70 and $r_{\rm late} \sim 0.60$–0.65.

(3) Given $L_{\rm UV,tot}/L_{3.6,\rm tot}$ is a good proxy for sSFR, it follows that both early- and late-type galaxies exhibit log-linear inverse correlations between $M_{\rm BH}$ and sSFR, the latter having a steeper dependence on sSFR (i.e., $M_{\rm BH} \propto {\rm sSFR}_{\rm FUV}^{-2.58}$) than the former ($M_{\rm BH} \propto {\rm sSFR}_{\rm FUV}^{-4.38}$). This suggests different channels for SMBH growth in early- and late-type galaxies.

(4) We have compared the $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ relations with the $M_{\rm BH} - \sigma$ and $M_{\rm BH} - L_{3.6,\rm tot}$ relations for our sample galaxies. While the $M_{\rm BH} - \mathcal{C}_{\rm UV}$ relations are marginally weaker ($r \sim 0.60$–0.70) and typically have 5%–27% more scatter than the $M_{\rm BH} - \sigma$ relations ($r \sim 0.72$–0.78), the former potentially constrains SMBH-galaxy coevolution models that predict different SMBH growth for different morphologies. In contrast, the $M - \sigma$ relations for late- and early-type galaxies are similar. The $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ and $M_{\rm BH} - L_{3.6,\rm tot}$ relations display a similar level of strength and vertical scatter.

(5) We did not detect departures (bends or offsets) from the $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ relations because of core-Sérsic or Sérsic galaxies. However, we cannot firmly rule out the presence of such substructures because our sample does not consist of a large number of core-Sérsic galaxies.

(6) We argue that the different $M_{\rm BH} - \mathcal{C}_{\rm UV}$ relations for early- and late-type galaxies reflect that the two Hubble types have two distinct SMBH feeding mechanisms. Massive early-type galaxies ($M_{*,k} \sim 8 \times 10^{10} M_\odot$–$10^{12} M_\odot$) at the high-mass end of the $M_{\rm BH} - \mathcal{C}_{\rm UV}$ red sequence are core-Sérsic galaxies. Their formation is consistent with the scenario in which a gas-rich (wet) major merger at high redshift drives intense bursts of star formation, efficient SMBH growth, and the ensuing rapid quenching of star formation by strong AGN feedback. This is accompanied by gas-poor (dry) major mergers since $z \sim 1.5$–2, involving a low level of star formation. In contrast, the less massive (Sérsic) early-type galaxies ($M_{*,k} \sim 10^{10} M_\odot - 2 \times 10^{11} M_\odot$) in the low-mass part of the $M_{\rm BH} - \mathcal{C}_{\rm UV}$ red sequence are likely built up primarily via gas-rich major mergers and form their stellar populations over an extended period of time. We tentatively hypothesize that late-type galaxies ($M_{*,k} \sim 10^9 M_\odot - 2 \times 10^{11} M_\odot$) that define the $M_{\rm BH} - \mathcal{C}_{\rm UV}$ blue sequence form via secular-driven processes involving nonaxisymmetric stellar structures, such as bars and spiral arms. Gas-rich minor mergers could also account for the build-up of late-type galaxies.

(7) Having demonstrated that our $M_{\rm BH} - \mathcal{C}_{\rm UV}$ relations are able to predict the SMBH masses in 10 galaxies, we employed these new relations to estimate the central black hole masses in 1382 GALEX/S^{4}G galaxies with no measured black hole masses after excluding highly inclined and dust-obscured GALEX/S^{4}G galaxies (Bouquin et al. 2018). We suggest that the $M_{\rm BH} - \mathcal{C}_{\rm UV}$ relations can be used to estimate black hole masses without the need for high-resolution spectroscopy. However, we warn that to do so, one should use galaxy colors that are determined based on UV and 3.6 μm magnitudes that have been obtained in a homogeneous way. Furthermore, the

[13] All the seven core-Sérsic galaxies in this paper are "normal-core" galaxies (Dullo 2019).

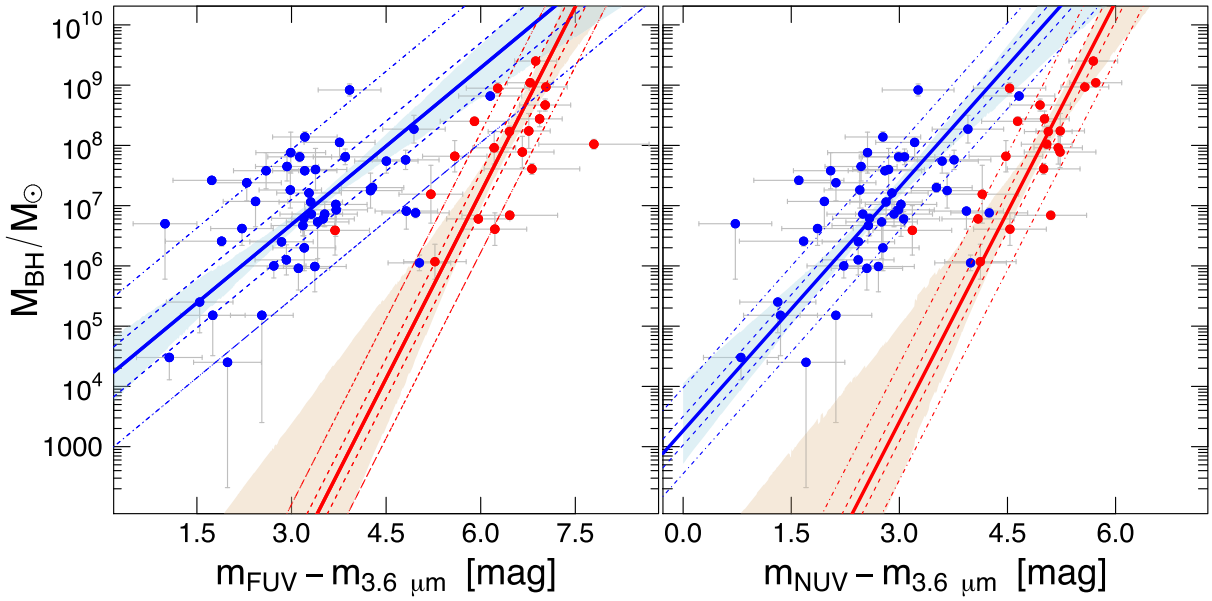

Figure A1. Similar to Figure 2, but here plotting the results of our symmetric linear Bayesian regression analysis. Early-type and late-type galaxies are shown in red and blue, respectively. The shaded regions indicate the 95% HDI for the derived fits. The dotted lines mark the 1σ and 3σ intervals for the additional real dispersion that is not explained by the error bars; see the text for further detail.

$M_{BH} - \mathcal{C}_{UV}$ relations can be used to identify low-mass and bulgeless galaxies that potentially harbor IMBHs.

We thank the referee for their useful comments. B.T.D. acknowledges support from the Spanish postdoctoral fellowship "Ayudas 1265 para la atracción del talento investigador. Modalidad 2: jóvenes investigadores." funded by Comunidad de Madrid under grant number 2016-T2/TIC-2039. B.T.D. acknowledges support from grant "Ayudas para la realización de proyectos de I+D para jóvenes doctores 2019." funded by Comunidad de Madrid and Universidad Complutense de Madrid under grant number PR65/19-22417. We acknowledge financial support from the Spanish Ministry of Economy and Competitiveness (MINECO) under grant numbers AYA2016-75808-R and RTI, which is partly funded by the European Regional Development Fund, and from the Excellence Network MaegNet (AYA2017-90589-REDT). A.Y.K.B. acknowledges financial support from the Spanish Ministry of Economy and Competitiveness (MINECO), project Estallidos AYA2016-79724-C4-2-P. J.H.K. acknowledges financial support from the European Union's Horizon 2020 research and innovation program under Marie Skłodowska-Curie grant agreement No 721463 to the SUNDIAL ITN network, from the State Research Agency (AEI) of the Spanish Ministry of Science, Innovation and Universities (MCIU) and the European Regional Development Fund (FEDER) under the grant with reference AYA2016-76219-P, from IAC project P/300724, financed by the Ministry of Science, Innovation and Universities, through the State Budget and by the Canary Islands Department of Economy, Knowledge and Employment, through the Regional Budget of the Autonomous Community, and from the Fundación BBVA under its 2017 program of assistance to scientific research groups, for the project "Using machine-learning techniques to drag galaxies from the noise in deep imaging." J.G. acknowledges financial support from the Spanish Ministry of Economy and Competitiveness under grant number AYA2016-77237-C3-2P.

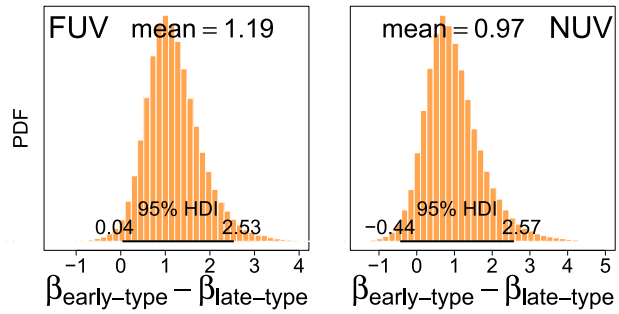

Figure A2. PDF for the difference in slopes (β) between the linear Bayesian regression fits to the early- and late-type (M_{BH}, $\mathcal{C}_{UV,tot}$) data (see Figure A1).

Appendix A

A.1. Bayesian Approach

We have performed linear regression fits to the (M_{BH}, $\mathcal{C}_{UV,tot}$) data set by applying a Bayesian statistical inference with an MCMC technique.[14] The regression fits take into account errors in M_{BH} and $\mathcal{C}_{UV,tot}$ and an additional spread that is not explained by the error bars. The MCMC runs were implemented in our work through the Stan programming language[15] (e.g., Carpenter et al. 2017). In addition to the slope, intercept, and dispersion, the model parameters include "true" color and SMBH mass values. In our implementation, the observed SMBH mass upper limits were modeled by drawing samples from an asymmetric normal distribution centered around the true values, with the standard deviation having an upper wing that equals the error bar of the measurements and a lower wing with a much larger size. The regression fits thus probe a large range in SMBH masses for galaxies with SMBH upper limits. For the Bayesian analysis, we used noninformative prior distributions (e.g., Gelman et al. 2013) and confirmed that the results did not depend on the choice of the prior details.

[14] Interested readers are referred to Andreon & Weaver (2015, page 134) and Hilbe et al. (2017, page 278) for a description of the MCMC method.
[15] https://mc-stan.org

Figure A1 shows the results of the Bayesian linear regressions. The shaded regions indicate the 95% highest density interval (HDI) for the derived fits. Note that the HDIs are not symmetric because the derived parameters exhibit skewed posterior distributions. The dotted lines mark the 1σ and 3σ intervals for the additional real dispersion that is not explained by the error bars (Figure A1). The probability distribution functions (PDFs) for these additional dispersions are not symmetrical, and the derived values exhibit 95% HDIs of 0.00 and 0.74, and 0.14 and 0.68 for the early- and late-type galaxies. For the former, the additional dispersion is not significantly different from zero, therefore the observed dispersion could be fully explained by the error bars. In contrast, an additional real dispersion is needed for the latter to explain the residuals from the fitted relation.

In Figure A2 we show the PDF for the disparity in slope between early- and late-type $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ relations obtained by applying the linear Bayesian regression fits. The PDFs are determined using all the individual steps of the computed chains, thus avoiding any assumption about the probability distribution of the derived parameters. We find that the significance levels for rejecting the null hypothesis that both morphological types have the same slope are 1.7% and 6.7% for the FUV and NUV relations, respectively.

Appendix B

B.1. Outliers in the $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ Diagrams

Five galaxies in our sample are offset notably from the $M_{\rm BH} - \mathcal{C}$ relations (Figure 2).

B.1.1. NGC 2685

The Helix Galaxy NGC 2685 is a well-studied polar ring lenticular galaxy (Schechter & Gunn 1978; Shane 1980; Watson et al. 1994; Eskridge & Pogge 1997; Schinnerer & Scoville 2002; Karataeva et al. 2004; Rampazzo et al. 2007; Józsa et al. 2009). Sandage (1961) referred to it as to the most unusual of all the galaxies in his atlas, and subsequent studies confirmed its rare nature (e.g., Eskridge & Pogge 1997; Schinnerer & Scoville 2002; Józsa et al. 2009). The galaxy contains a high concentration of molecular, neutral, and atomic hydrogen gas in its polar ring. Figure 2 shows that it resides in the $M_{\rm BH} - \mathcal{C}$ blue sequence that is defined by late-type galaxies, and its $\mathcal{C}_{\rm UV,tot}$ color is abnormally ($\sim$1.90/1.00 FUV/NUV mag) bluer than that predicted for early-type galaxies based on the galaxy SMBH mass. This result agrees with the two formation scenarios considered in the literature for NGC 2685: formation via accretion of a small gas-rich companion (e.g., Shane 1980; Watson et al. 1994; Schinnerer & Scoville 2002), and a merger of two disk galaxies (e.g., Józsa et al. 2009).

B.1.2. NGC 3310

The starburst galaxy NGC 3310 is a well-studied peculiar spiral galaxy (SAB) known for its very blue color, circumnuclear ring of star formation, tidal features, and very bright infrared luminosity (e.g., Balick & Heckman 1981; Mulder et al. 1995; Smith et al. 1996; Conselice et al. 2000; Elmegreen et al. 2002; Wehner & Gallagher 2005; Wehner et al. 2006; Hägele et al. 2010). The favored scenario for the formation of the galaxy is through accretion of a small gas-rich dwarf galaxy $\sim$10 Myr ago (Balick & Heckman 1981; Mulder et al. 1995; Smith et al. 1996; Conselice et al. 2000; Wehner & Gallagher 2005; Wehner et al. 2006). Consistent with this picture, Figure 2 shows that the galaxy $\mathcal{C}_{\rm UV,tot}$ color is ($\sim$1.99/1.79 FUV/NUV mag) bluer than expected for late-type galaxies.

B.1.3. NGC 3368

The case of the double-barred spiral (SAB) galaxy NGC 3368 is less clear. It is the brightest galaxy in the nearby NGC 3368 group (Sil'chenko et al. 2003; Watkins et al. 2014), containing a dominant pseudo-bulge, a small classical bulge, and a box/peanut component (Nowak et al. 2010; Erwin et al. 2015). Not only is its $\mathcal{C}_{\rm UV,tot}$ color ($\sim$1.81/1.61 FUV/NUV mag) redder than that expected for late-type galaxies given its $M_{\rm BH}$, the galaxy also falls on the $M_{\rm BH} - \mathcal{C}_{\rm NUV,tot}$ red sequence that is defined by early-type galaxies (Figure 2). Figure 2 shows that NGC 3368 lies 0.45 dex below the best-fitting $M_{\rm BH} - \sigma$ line in the log $M_{\rm BH}$ direction. This might suggest that the galaxy has an abnormally low SMBH mass and not a discrepant redder color. We cannot rule out the possibility that the black hole mass and color conspire, causing the offset in the $M_{\rm BH} - \mathcal{C}$ diagrams (Figure 2).

B.1.4. NGC 4826

The SABa galaxy NGC 4826 is also referred to as the Evil- or Black-Eye Galaxy, which refers to its dusty disk, which has a radius $R \sim 50''$. It has a large-scale counter-rotating gas disk (van Driel & Buta 1993; Braun et al. 1994; García-Burillo et al. 2003). The galaxy is the most significant outlier in the $M_{\rm BH} - \mathcal{C}_{\rm UV,tot}$ diagrams, having a $\mathcal{C}_{\rm UV,tot}$ color ($\sim$2.68/2.04 FUV/NUV mag) that is redder than what is expected for late-type galaxies (Figure 2). The dust disk likely explains the deviant (redder) UV $-$[3.6] color for the galaxy. Figure 2 shows that the galaxy falls on the red early-type sequence, despite being a late-type galaxy.

B.1.5. NGC 5018

The giant elliptical (E3) NGC 5018 is the most intriguing case. It is the brightest member of the poor NGC 5018 group that contains five galaxies (Gourgoulhon et al. 1992) and is known for its morphological peculiarity, including shells, ripples, and a dust lane (Schweizer 1987; Kim et al. 1988; Malin & Hadley 1997; Leonardi & Worthey 2000; Knapen et al. 2014). There is a tidal bridge between NGC 5018 and a gas-rich spiral companion, NGC 5022, detected in optical (Schweizer 1987; Malin & Hadley 1997) and H I observations (Kim et al. 1988; Ghosh et al. 2005) that was built up via accretion of gaseous material from NGC 5022 onto NGC 5018. NGC 5018 is currently forming stars (Ghosh et al. 2005), and about 12% (by mass) of the stars in the galaxy are younger than 3.4 Gyr (Nolan et al. 2007, see also Leonardi & Worthey 2000). The galaxy has a low metallicity and FUV and mid-UV flux deficits that are unusual for its luminosity and velocity dispersion (Bertola et al. 1993; Buson et al. 2004). While Carollo & Danziger (1994) attributed the observed unusual metallicity and UV fluxes to nuclear dust extinction, the galaxy is an outlier from the $\mathcal{C}_{\rm FUV,tot} - M_{\rm BH}$ diagram but not from the $\mathcal{C}_{\rm NUV,tot} - M_{\rm BH}$ diagram (Figure 2), revealing that the offset nature of the galaxy is not caused by dust extinction, but rather by the mixing of young and old populations of stars at the galaxy center, which washes out the preexisting metallicity (Bertola et al. 1993; Hilker & Kissler-Patig 1996; Kim et al. 1988; Buson et al. 2004). We also note that the galaxy is slightly offset from the $M_{\rm BH} - L_{3.6,\rm tot}$ relation toward a brighter magnitude. This may partly explain the deviant red color of the galaxy.

Appendix C

Table C1 presents total magnitudes, bulge-to-total (B/T) and disk-to-total (D/T) ratios, dust corrections, and SMBH masses for our sample galaxies.

Table C1
Basic Data for Our Sample of 67 GALEX/S^{4}G Galaxies with Directly Measured SMBH Masses

Galaxy (1)	Type (2)	m_{FUV} B+D (mag) (3)	m_{NUV} B+D (mag) (4)	$m_{3.6}$ B+D (mag) (5)	B/T FUV and NUV (6)	B/T 3.6 (7)	D/T FUV and NUV (8)	D/T 3.6 (9)	$Dust_{corr}$ (B and D)$_{3.6}$ (mag) (10)	$Dust_{corr}$ (B and D)$_{FUV}$ (mag) (11)	$Dust_{corr}$ (B and D)$_{NUV}$ (mag) (12)	log M_{BH} ($M_\odot$) (13)
NGC 0289	SBbc	$12.84^{+0.39}_{-0.39}$	$12.67^{+0.40}_{-0.40}$	$10.55^{+0.38}_{-0.38}$	⋯	0.02	0.92	0.92	0.14/0.05	⋯/0.66	⋯/0.66	$7.38^{+0.30}_{-7.38}$
NGC 0428	SABm	$12.91^{+0.37}_{-0.37}$	$12.65^{+0.37}_{-0.37}$	$11.85^{+0.35}_{-0.35}$	⋯	0.00	0.93	0.93	0.13/0.05	⋯/0.70	⋯/0.70	$4.48^{-0.37}_{-4.48}$
NGC 0613	SBbc	$12.99^{+0.36}_{-0.36}$	$12.46^{+0.36}_{-0.36}$	$9.61^{+0.34}_{-0.34}$	0.04	0.13	0.92	0.75	0.13/0.05	1.29/0.62	1.29/0.62	$7.60^{+0.35}_{-0.35}$
NGC 1042	SABc	$13.08^{+0.39}_{-0.39}$	$12.80^{+0.39}_{-0.39}$	$11.10^{+0.37}_{-0.37}$	⋯	0.02	0.97	0.97	0.13/0.05	⋯/0.70	⋯/0.70	$4.40^{+2.08}_{-4.40}$
NGC 1052	E4	$16.81^{+0.24}_{-0.24}$	$15.28^{+0.24}_{-0.24}$	$10.05^{+0.23}_{-0.23}$	0.67	0.72	⋯	⋯	0.01/⋯	0.23/⋯	0.23/⋯	$8.24^{+0.29}_{-0.29}$
NGC 1097	SBb	$12.10^{+0.37}_{-0.37}$	$11.66^{+0.37}_{-0.37}$	$8.89^{+0.35}_{-0.35}$	0.11	0.22	0.79	0.56	0.15/0.05	1.41/0.73	1.41/0.73	$8.14^{+0.09}_{-0.09}$
NGC 1300	SBc	$13.21^{+0.40}_{-0.40}$	$12.78^{+0.40}_{-0.40}$	$10.23^{+0.38}_{-0.38}$	⋯	0.07	0.87	0.87	0.16/0.05	⋯/0.76	⋯/0.76	$7.88^{+0.34}_{-0.34}$
NGC 1386	SB0-a	$15.79^{+0.35}_{-0.35}$	$14.64^{+0.36}_{-0.36}$	$10.51^{+0.32}_{-0.32}$	0.26	0.36	0.70	0.58	0.32/0.10	1.54/1.00	1.54/1.00	$6.07^{+0.29}_{-0.29}$
NGC 1493	SBc	$12.99^{+0.40}_{-0.40}$	$12.76^{+0.38}_{-0.38}$	$11.45^{+0.36}_{-0.36}$	⋯	0.00	0.97	0.97	0.11/0.04	⋯/0.53	⋯/0.53	$5.40^{+0.51}_{-5.40}$
NGC 2685	SB0-a	$14.66^{+0.39}_{-0.39}$	$14.15^{+0.39}_{-0.39}$	$10.97^{+0.36}_{-0.36}$	0.36	0.46	0.54	0.41	0.23/0.08	1.82/1.17	1.83/1.18	$6.59^{+0.41}_{-6.59}$
NGC 2748	Sbc	$13.98^{+0.36}_{-0.36}$	$13.52^{+0.36}_{-0.36}$	$11.05^{+0.34}_{-0.34}$	⋯	0.03	0.97	0.97	0.34/0.11	⋯/1.16	⋯/1.17	$7.65^{+0.24}_{-0.24}$
NGC 2787	SB0-a	$16.66^{+0.54}_{-0.54}$	$14.85^{+0.36}_{-0.36}$	$9.85^{+0.35}_{-0.35}$	0.39	0.51	0.59	0.45	0.19/0.07	2.32/1.65	2.34/1.66	$7.61^{+0.09}_{-0.09}$
NGC 2903	SBbc	$11.66^{+0.33}_{-0.33}$	$11.17^{+0.36}_{-0.36}$	$8.35^{+0.34}_{-0.34}$	⋯	0.07	0.90	0.90	0.25/0.08	⋯/0.96	⋯/0.96	$7.06^{+0.28}_{-7.06}$
NGC 2964	SBbc	$14.13^{+0.35}_{-0.35}$	$13.47^{+0.35}_{-0.35}$	$10.72^{+0.34}_{-0.34}$	⋯	0.10	0.90	090	0.20/0.06	⋯/0.75	⋯/0.75	$6.73^{+0.61}_{-6.73}$
NGC 2974	E4	$16.98^{+0.23}_{-0.23}$	$15.59^{+0.24}_{-0.24}$	$10.52^{+0.22}_{-0.22}$	0.49	0.54	⋯	⋯	0.01/⋯	0.44/⋯	0.45/0.45	$8.23^{+0.09}_{-0.09}$
NGC 3021	Sbc	$14.60^{+0.35}_{-0.35}$	$14.07^{+0.35}_{-0.35}$	$11.62^{+0.34}_{-0.34}$	⋯	0.01	0.98	0.98	0.19/0.06	⋯/0.71	⋯/0.71	$7.26^{+0.30}_{-7.26}$
NGC 3031	Sab	$10.07^{+0.36}_{-0.36}$	$9.29^{+0.45}_{-0.45}$	$6.22^{+0.39}_{-0.39}$	0.14	0.46	0.86	0.54	0.23/0.08	1.96/1.30	1.97/1.31	$7.81^{+0.13}_{-0.13}$
NGC 3079	SBc	$12.10^{+0.36}_{-0.36}$	$11.69^{+0.36}_{-0.36}$	$9.26^{+0.34}_{-0.34}$	0.07	0.24	0.93	0.77	0.57/0.24	1.81/1.73	1.82/1.73	$6.40^{+0.05}_{-0.05}$
NGC 3115	E-S0	$14.45^{+0.36}_{-0.36}$	$12.71^{+0.35}_{-0.35}$	$8.18^{+0.33}_{-0.33}$	0.63	0.74	0.37	0.26	0.36/0.13	1.86/1.37	1.87/1.38	$8.95^{+0.09}_{-0.09}$
NGC 3310	SBbc	$11.68^{+0.36}_{-0.36}$	$11.41^{+0.36}_{-0.36}$	$10.69^{+0.34}_{-0.34}$	0.16	0.33	0.83	0.63	0.13/0.05	1.33/0.65	1.33/0.66	$6.70^{+0.92}_{-6.70}$
NGC 3368	SBab	$13.76^{+0.38}_{-0.38}$	$13.04^{+0.38}_{-0.38}$	$8.79^{+0.36}_{-0.36}$	⋯	0.07	0.92	0.92	0.15/0.05	⋯/0.71	⋯/0.71	$6.88^{+0.08}_{-0.08}$
NGC 3414	SB0	$16.39^{+0.39}_{-0.39}$	$15.13^{+0.36}_{-0.36}$	$10.49^{+0.34}_{-0.34}$	0.57	0.69	0.41	0.29	0.13/0.05	1.36/0.69	1.37/0.69	$8.40^{+0.07}_{-0.07}$
NGC 3423	Sc	$12.92^{+0.37}_{-0.37}$	$12.52^{+0.37}_{-0.37}$	$11.17^{+0.35}_{-0.35}$	⋯	0.06	0.95	0.95	0.12/0.05	⋯/0.71	⋯/0.71	$5.18^{+0.67}_{-5.18}$
NGC 3489	SB0-a	$16.01^{+0.36}_{-0.36}$	$14.14^{+0.36}_{-0.36}$	$10.05^{+0.33}_{-0.33}$	0.23	0.32	0.71	0.58	0.19/0.06	1.38/0.71	1.39/0.71	$6.78^{+0.05}_{-0.05}$
NGC 3608	E2	$17.58^{+0.29}_{-0.29}$	$15.51^{+0.29}_{-0.29}$	$10.56^{+0.27}_{-0.27}$	1.00	1.00	⋯	⋯	0.01/⋯	0.17/⋯	0.18/⋯	$8.67^{+0.10}_{-0.10}$
NGC 3627	SBb	$12.07^{+0.35}_{-0.35}$	$11.35^{+0.35}_{-0.35}$	$8.36^{+0.34}_{-0.34}$	0.03	0.12	0.93	0.78	0.26/0.08	1.62/1.00	1.62/1.00	$6.93^{+0.05}_{-0.05}$
NGC 3642	Sbc	$13.25^{+0.44}_{-0.44}$	$13.12^{+0.39}_{-0.39}$	$11.51^{+0.42}_{-0.42}$	0.03	0.14	0.95	0.80	0.12/0.04	1.21/0.55	1.21/0.55	$7.42^{+0.04}_{-7.42}$
NGC 4041	Sbc	$13.45^{+0.36}_{-0.36}$	$12.96^{+0.36}_{-0.36}$	$10.73^{+0.35}_{-0.35}$	0.03	0.14	0.95	0.80	0.12/0.04	1.28/0.61	1.28/0.61	$6.00^{+0.20}_{-6.00}$
NGC 4051	SABb	$12.89^{+0.36}_{-0.36}$	$12.40^{+0.36}_{-0.36}$	$9.97^{+0.34}_{-0.34}$	0.03	0.15	0.96	0.80	0.13/0.04	1.26/0.59	1.27/0.59	$6.10^{+0.25}_{-0.25}$
NGC 4088	SABc	$12.92^{+0.37}_{-0.37}$	$12.31^{+0.37}_{-0.37}$	$9.72^{+0.35}_{-0.35}$	⋯	0.02	0.98	0.98	0.32/0.11	⋯/1.08	⋯/1.08	$6.79^{+0.29}_{-6.79}$
NGC 4151	SBab	$13.16^{+0.36}_{-0.36}$	$13.02^{+0.36}_{-0.36}$	$10.03^{+0.35}_{-0.35}$	0.22	0.44	0.78	0.56	0.14/0.05	1.39/0.71	1.39/0.71	$7.81^{+0.08}_{-0.08}$
NGC 4203	E-S0	$15.55^{+0.37}_{-0.37}$	$14.44^{+0.36}_{-0.36}$	$9.96^{+0.33}_{-0.33}$	1.0	1.0	⋯	⋯	0.11/0.04	1.20/⋯	1.20/⋯	$7.82^{+0.26}_{-0.26}$
NGC 4212	Sc	$14.08^{+0.36}_{-0.36}$	$13.42^{+0.36}_{-0.36}$	$10.71^{+0.34}_{-0.34}$	⋯	0.04	0.96	0.96	0.18/0.06	⋯/0.82	⋯/0.82	$5.99^{+0.42}_{-5.99}$
NGC 4245	SB0-a	$16.20^{+0.36}_{-0.36}$	$15.14^{+0.37}_{-0.37}$	$10.99^{+0.34}_{-0.34}$	0.11	0.18	0.82	0.70	0.13/0.05	1.32/0.64	1.32/0.64	$7.19^{+0.48}_{-7.19}$
NGC 4258	SBbc	$11.10^{+0.36}_{-0.36}$	$10.69^{+0.36}_{-0.36}$	$7.89^{+0.35}_{-0.35}$	⋯	0.07	0.96	0.96	0.32/0.10	⋯/1.02	⋯/1.02	$7.58^{+0.03}_{-0.03}$
NGC 4278	E1-2	$16.00^{+0.24}_{-0.24}$	$14.99^{+0.24}_{-0.24}$	$9.79^{+0.22}_{-0.22}$	0.65	0.71	⋯	⋯	0.01/⋯	0.25/⋯	0.26/⋯	$7.96^{+0.27}_{-0.27}$
NGC 4314	SBa	$15.16^{+0.35}_{-0.35}$	$14.27^{+0.35}_{-0.35}$	$10.34^{+0.34}_{-0.34}$	0.08	0.17	0.79	0.57	0.12/0.05	1.32/0.67	1.33/0.67	$6.91^{+0.30}_{-6.91}$
NGC 4321	SABb	$12.17^{+0.36}_{-0.36}$	$11.56^{+0.36}_{-0.36}$	$8.99^{+0.34}_{-0.34}$	⋯	0.10	0.90	0.90	0.12/0.05	⋯/0.67	⋯/0.67	$6.67^{+0.17}_{-6.67}$
NGC 4371	SB0-a	$16.90^{+0.54}_{-0.54}$	$15.54^{+0.36}_{-0.36}$	$10.44^{+0.32}_{-0.32}$	0.16	0.23	0.74	0.62	0.12/0.05	1.41/0.76	1.41/0.76	$6.84^{+0.07}_{-0.07}$
NGC 4374	E1	$15.77^{+0.24}_{-0.24}$	$14.31^{+0.23}_{-0.23}$	$8.74^{+0.22}_{-0.22}$	0.60	0.65	⋯	⋯	0.01/⋯	0.35/⋯	0.35/⋯	$8.97^{+0.05}_{-0.05}$
NGC 4388	SBb	$13.42^{+0.35}_{-0.35}$	$12.60^{+0.36}_{-0.36}$	$10.11^{+0.34}_{-0.34}$	0.06	0.16	0.94	0.84	0.56/0.24	1.97/1.84	1.97/1.85	$6.86^{+0.04}_{-0.04}$
NGC 4472	E2	$14.78^{+0.21}_{-0.21}$	$13.60^{+0.22}_{-0.22}$	$7.91^{+0.20}_{-0.20}$	0.58	0.63	⋯	⋯	0.01/⋯	0.19/⋯	0.19/⋯	$9.40^{+0.04}_{-0.04}$
NGC 4501	Sb	$13.04^{+0.35}_{-0.35}$	$12.27^{+0.35}_{-0.35}$	$8.76^{+0.39}_{-0.39}$	⋯	0.06	0.94	0.94	0.21/0.07	⋯/0.92	⋯/0.93	$7.30^{+0.08}_{-0.08}$
NGC 4548	SBb	$14.07^{+0.37}_{-0.37}$	$13.48^{+0.36}_{-0.36}$	$9.82^{+0.35}_{-0.35}$	0.03	0.12	0.93	0.78	0.13/0.05	1.45/0.78	1.46/0.79	$7.25^{+0.29}_{-0.29}$
NGC 4593	SBb	$14.26^{+0.37}_{-0.37}$	$13.66^{+0.38}_{-0.38}$	$10.74^{+0.36}_{-0.36}$	0.10	0.16	0.81	0.61	0.13/0.05	1.36/0.69	1.37/0.69	$6.86^{+0.21}_{-0.21}$
NGC 4594	Sa	$13.35^{+0.36}_{-0.36}$	$11.86^{+0.37}_{-0.37}$	$7.20^{+0.35}_{-0.35}$	0.49	0.83	0.51	0.17	0.31/0.10	1.84/1.27	1.84/1.28	$8.82^{+0.05}_{-0.05}$
NGC 4596	SB0-a	$16.72^{+0.36}_{-0.36}$	$15.29^{+0.35}_{-0.35}$	$10.06^{+0.32}_{-0.32}$	0.13	0.19	0.77	0.65	0.13/0.05	1.32/0.64	1.32/0.65	$7.89^{+0.26}_{-0.26}$
NGC 4698	Sab	$14.89^{+0.38}_{-0.38}$	$13.84^{+0.36}_{-0.36}$	$10.08^{+0.35}_{-0.35}$	0.22	0.44	0.78	0.56	0.17/0.05	1.43/0.74	1.43/0.75	$7.76^{+0.16}_{-0.16}$

Table C1
(Continued)

Galaxy (1)	Type (2)	$m_{\rm FUV}$ B+D (mag) (3)	$m_{\rm NUV}$ B+D (mag) (4)	$m_{3.6}$ B+D (mag) (5)	B/T FUV and NUV (6)	B/T 3.6 (7)	D/T FUV and NUV (8)	D/T 3.6 (9)	$\rm Dust_{corr}$ (B and D)$_{3.6}$ (mag) (10)	$\rm Dust_{corr}$ (B and D)$_{\rm FUV}$ (mag) (11)	$\rm Dust_{corr}$ (B and D)$_{\rm NUV}$ (mag) (12)	log $M_{\rm BH}$ ($M_\odot$) (13)
NGC 4736	Sab	$11.51^{+0.36}_{-0.36}$	$11.07^{+0.36}_{-0.36}$	$8.01^{+0.34}_{-0.34}$	0.24	0.42	0.54	0.21	0.12/0.04	1.27/0.61	1.28/0.61	$6.78^{+0.12}_{-0.12}$
NGC 4800	Sb	$14.64^{+0.34}_{-0.34}$	$13.96^{+0.35}_{-0.35}$	$10.94^{+0.34}_{-0.34}$	0.04	0.17	0.95	0.80	0.13/0.05	1.28/0.60	1.28/0.61	$7.02^{+0.53}_{-7.02}$
NGC 4826	SABa	$12.94^{+0.36}_{-0.36}$	$11.90^{+0.36}_{-0.36}$	$7.91^{+0.34}_{-0.34}$	0.12	0.28	0.87	0.68	0.21/0.07	1.62/0.96	1.63/0.96	$6.05^{+0.13}_{-0.13}$
NGC 5005	SABb	$14.11^{+0.35}_{-0.35}$	$13.12^{+0.35}_{-0.35}$	$9.17^{+0.33}_{-0.33}$	0.06	0.15	0.88	0.69	0.24/0.07	1.44/0.81	1.45/0.81	$8.27^{+0.23}_{-0.23}$
NGC 5018	E3	$18.07^{+0.63}_{-0.63}$	$15.32^{+0.25}_{-0.25}$	$10.28^{+0.27}_{-0.27}$	0.83	0.88	⋯	⋯	0.02/⋯	0.79/⋯	0.80/⋯	$8.02^{+0.08}_{-0.08}$
NGC 5055	Sbc	$11.89^{+0.36}_{-0.36}$	$11.23^{+0.36}_{-0.36}$	$7.97^{+0.34}_{-0.34}$	0.03	0.17	0.97	0.81	0.19/0.06	1.40/0.73	1.40/0.73	$8.92^{+0.10}_{-0.10}$
NGC 5194	SABb	$10.60^{+0.35}_{-0.35}$	$10.04^{+0.34}_{-0.34}$	$7.49^{+0.32}_{-0.32}$	⋯	0.10	0.87	0.88	0.17/0.06	⋯/0.83	⋯/0.83	$5.96^{+0.36}_{-5.96}$
NGC 5248	SBb	$12.86^{+0.36}_{-0.36}$	$12.43^{+0.36}_{-0.36}$	$9.65^{+0.33}_{-0.33}$	0.05	0.20	0.95	0.80	0.14/0.05	1.37/0.69	1.37/0.69	$6.30^{+0.38}_{-0.38}$
NGC 5273	S0	$17.46^{+0.36}_{-0.36}$	$15.77^{+0.36}_{-0.36}$	$11.24^{+0.34}_{-0.34}$	⋯	0.09	0.99	0.91	0.11/0.04	⋯/0.53	⋯/0.53	$6.61^{+0.27}_{-0.27}$
NGC 5347	SBab	$15.27^{+0.37}_{-0.37}$	$14.89^{+0.38}_{-0.38}$	$11.99^{+0.34}_{-0.34}$	0.17	0.33	0.80	0.59	0.13/0.05	1.31/0.64	1.32/0.64	$7.21^{+0.42}_{-7.21}$
NGC 5427	SABc	$13.59^{+0.37}_{-0.37}$	$13.04^{+0.37}_{-0.37}$	$10.99^{+0.35}_{-0.35}$	⋯	0.10	0.90	0.90	0.12/0.05	⋯/0.68	⋯/0.69	$7.58^{+0.30}_{-7.58}$
NGC 5457	SABc	$9.65^{+0.64}_{-0.64}$	$9.43^{+0.63}_{-0.63}$	$7.76^{+0.61}_{-0.61}$	⋯	0.05	0.95	0.95	0.11/0.04	⋯/0.53	⋯/0.53	$6.41^{+0.08}_{-6.41}$
NGC 5576	E3	$17.35^{+0.26}_{-0.26}$	$15.43^{+0.24}_{-0.24}$	$10.42^{+0.23}_{-0.23}$	0.69	0.74	⋯	⋯	0.01/⋯	0.25/⋯	0.26/⋯	$8.44^{+0.13}_{-0.13}$
NGC 5728	SBa	$14.75^{+0.40}_{-0.40}$	$14.20^{+0.38}_{-0.38}$	$10.99^{+0.38}_{-0.38}$	0.12	0.26	0.72	0.39	0.20/0.07	2.09/1.41	2.10/1.42	$8.05^{+0.29}_{-0.29}$
NGC 5846	E0-1	$16.24^{+0.26}_{-0.26}$	$15.18^{+0.25}_{-0.25}$	$9.46^{+0.21}_{-0.21}$	0.60	0.65	⋯	⋯	0.01/⋯	0.46/⋯	0.46/⋯	$9.04^{+0.06}_{-0.06}$
NGC 5879	SBbc	$13.14^{+0.35}_{-0.35}$	$12.79^{+0.35}_{-0.35}$	$10.93^{+0.34}_{-0.34}$	0.16	0.46	0.83	0.51	0.44/0.16	1.67/1.33	1.67/1.33	$6.62^{+0.28}_{-6.62}$
NGC 5921	SBbc	$13.23^{+0.37}_{-0.37}$	$12.76^{+0.37}_{-0.37}$	$10.80^{+0.35}_{-0.35}$	0.01	0.11	0.99	0.83	0.13/0.05	1.45/0.78	1.45/0.79	$7.07^{+0.42}_{-7.07}$
NGC 7418	SBc	$13.35^{+0.36}_{-0.36}$	$12.94^{+0.35}_{-0.35}$	$10.82^{+0.33}_{-0.33}$	⋯	0.02	0.97	0.97	0.13/0.04	⋯/0.61	⋯/0.61	$5.18^{+1.78}_{-5.18}$
NGC 7582	SBab	$14.48^{+0.36}_{-0.36}$	$13.57^{+0.36}_{-0.36}$	$9.97^{+0.34}_{-0.34}$	0.10	0.25	0.78	0.45	0.29/0.09	1.51/0.93	1.51/0.93	$7.74^{+0.20}_{-0.20}$

Note. Column (1) lists the galaxy name. Column (2) lists the morphological type from HyperLeda, in good agreement with the classification in the NED database. Columns (3)–(5) list dust-corrected, FUV, NUV, and 3.6 μm total ($B + D$) apparent magnitudes ($m_{\rm FUV}$, $m_{\rm NUV}$, and $m_{3.6}$). Columns (6)–(9) list FUV, NUV, and 3.6 μm bulge-to-total (B/T) and disk-to-total (D/T) ratios. Columns (10)–(12) list bulge and disk dust corrections in $m_{3.6}$, FUV-, and NUV-bands, see Section 3.1. Column (13) list the SMBH mass from van den Bosch (2016).

(This table is available in machine-readable form.)

Appendix D

We use our $M_{\rm BH} - \mathcal{C}_{\rm FUV,tot}$ and $M_{\rm BH} - \mathcal{C}_{\rm NUV,tot}$ relations (Table 1) together with the appropriate asymptotic galaxy colors derived based on the asymptotic FUV, NUV, and 3.6 μm magnitudes from Bouquin et al. (2018, their Table 1) to predict tentative black hole masses in a sample of 1382 GALEX/S^{4}G galaxies with no measured black hole masses (Table D1). From the Bouquin et al. (2018) sample, we excluded galaxies that are highly inclined, dust obscured, and have prominent large-scale bars and rings. In contrast to the total (B+D) magnitudes in this paper (Table C1), the Bouquin et al. (2018) asymptotic magnitudes, which were not corrected for internal dust attenuation, contain additional fluxes from bars, rings, and nuclear components. We therefore caution about overinterpreting these predicted black hole masses. Furthermore, Chandra X-ray data or/and high-resolution radio data are important to confirm the presence of a central black hole in the low-mass ($M_* \lesssim 10^{10} M_\odot$) GALEX/S^{4}G galaxies.

Table D1
Predicted Black Hole Masses for 1382 GALEX/S^{4}G Galaxies with no Measured Black Hole Masses

Galaxy (1)	Type (2)	M_{BH} (FUV) (3)	M_{BH} (NUV) (4)	Galaxy (1)	Type (2)	M_{BH} (FUV) (3)	M_{BH} (NUV) (4)	Galaxy (1)	Type (2)	M_{BH} (FUV) (3)	M_{BH} (NUV) (4)	Galaxy (1)	Type (2)	M_{BH} (FUV) (3)	M_{BH} (NUV) (4)
ESO 013-016	7.5	5.35	5.42	ESO 159-025	10.0	4.84	4.37	ESO 286-044	−0.8	6.33	6.25	ESO 342-050	5.0	6.80	6.99
ESO 026-001	5.9	5.03	4.96	ESO 187-035	9.0	4.78	4.57	ESO 287-037	8.5	5.61	5.64	ESO 345-046	7.0	5.16	5.11
ESO 027-001	5.0	6.42	6.45	ESO 187-051	9.0	4.79	4.52	ESO 288-013	9.0	5.46	5.26	ESO 347-008	9.0	4.50	4.16
ESO 027-008	5.1	7.98	8.15	ESO 202-041	8.9	3.86	3.34	ESO 289-026	7.9	4.98	4.72	ESO 347-017	9.0	4.83	4.52
ESO 048-017	6.9	5.10	4.86	ESO 234-043	8.9	5.19	4.75	ESO 291-024	5.0	5.57	5.40	ESO 347-029	7.9	5.02	4.78
ESO 079-005	7.0	5.08	4.75	ESO 234-049	4.1	5.28	5.00	ESO 293-034	6.2	6.89	7.06	ESO 355-026	4.2	6.13	6.16
ESO 079-007	4.0	5.32	5.12	ESO 237-052	7.5	5.27	5.06	ESO 293-045	7.8	4.14	3.69	ESO 356-018	5.0	5.36	5.24
ESO 085-014	9.0	4.75	4.08	ESO 238-018	5.5	4.62	4.35	ESO 298-015	6.2	5.50	5.33	ESO 357-007	9.0	5.18	4.93
ESO 085-047	9.0	3.81	3.19	ESO 245-005	9.9	3.64	3.04	ESO 298-023	6.0	4.68	4.35	ESO 357-012	7.0	5.03	4.83
ESO 119-016	9.7	5.12	4.82	ESO 245-007	9.5	6.22	5.17	ESO 300-014	8.9	5.68	5.50	ESO 358-005	9.0	5.56	5.40
ESO 120-012	7.1	4.43	4.03	ESO 248-002	6.9	6.34	6.26	ESO 302-021	5.0	4.43	4.08	ESO 358-015	8.9	5.86	5.70
ESO 145-025	9.0	4.08	3.69	ESO 249-008	1.3	5.37	5.24	ESO 305-009	8.0	4.27	3.74	ESO 358-020	9.3	6.69	6.60
ESO 149-001	8.0	4.95	4.39	ESO 249-026	7.2	4.07	3.44	ESO 305-017	9.9	4.73	4.31	ESO 358-025	−2.6	5.05	5.35
ESO 149-003	9.7	3.56	3.07	ESO 249-035	5.9	4.28	3.68	ESO 340-017	8.0	5.50	5.72	ESO 358-051	0.0	6.33	6.23
ESO 150-005	7.8	5.07	4.86	ESO 249-036	10.0	4.66	4.31	ESO 340-042	7.8	5.31	5.41	ESO 358-054	8.0	5.02	4.79
ESO 154-023	8.9	4.65	4.75	ESO 285-048	5.9	5.73	5.55	ESO 341-032	9.0	5.18	4.92	ESO 358-060	9.9	3.74	3.30

Note. Column (1) lists the galaxy name. Column (2) lists the galaxy T-type from HyperLeda. Columns (3) and (4) list the tentative black hole masses (log $M_{BH}/M_{\odot}$) estimated using our $M_{BH} - \mathcal{C}_{FUV,tot}$ and $M_{BH} - \mathcal{C}_{NUV,tot}$ relations (Table 1) and the asymptotic FUV, NUV, and 3.6 μm magnitudes from Bouquin et al. (2018, their Table 1). We adopt a typical uncertainty of 0.85 dex on log(M_{BH}) for these predicted black hole masses.

(This table is available in its entirety in machine-readable form.)

ORCID iDs

Bililign T. Dullo https://orcid.org/0000-0002-4140-0110
Alexandre Y. K. Bouquin https://orcid.org/0000-0001-8285-4189
Armando Gil de Paz https://orcid.org/0000-0001-6150-2854
Johan H. Knapen https://orcid.org/0000-0003-1643-0024
Javier Gorgas https://orcid.org/0000-0001-7859-3676

References

Akritas, M. G., & Bershady, M. A. 1996, ApJ, 470, 706
Alam, S., Albareti, F. D., Allende Prieto, C., et al. 2015, ApJS, 219, 12
Alexander, D. M., Smail, I., Bauer, F. E., et al. 2005, Natur, 434, 738
Andreon, S., & Weaver, B. 2015, Bayesian Methods for the Physical Sciences. Learning from Examples in Astronomy and Physics (Berlin: Springer)
Athanassoula, E. 2005, MNRAS, 358, 1477
Bahcall, J. N., Kozlovsky, B.-Z., & Salpeter, E. E. 1972, ApJ, 171, 467
Baldassare, V. F., Reines, A. E., Gallo, E., & Greene, J. E. 2017, ApJ, 850, 196
Baldry, I. K., Glazebrook, K., Brinkmann, J., et al. 2004, ApJ, 600, 681
Balick, B., & Heckman, T. 1981, A&A, 96, 271
Barnes, J. E., & Hernquist, L. 1996, ApJ, 471, 115
Barnes, J. E., & Hernquist, L. E. 1991, ApJL, 370, L65
Barrows, R. S., Comerford, J. M., Zakamska, N. L., & Cooper, M. C. 2017, ApJ, 850, 27
Begelman, M. C., Blandford, R. D., & Rees, M. J. 1980, Natur, 287, 307
Beifiori, A., Courteau, S., Corsini, E. M., & Zhu, Y. 2012, MNRAS, 419, 2497
Bell, E. F., Naab, T., McIntosh, D. H., et al. 2006, ApJ, 640, 241
Bell, E. F., Wolf, C., Meisenheimer, K., et al. 2004, ApJ, 608, 752
Bertola, F., Burstein, D., & Buson, L. M. 1993, ApJ, 403, 573
Boselli, A., Cortese, L., Deharveng, J. M., et al. 2005, ApJL, 629, L29
Bouquin, A. Y. K., Gil de Paz, A., Boissier, S., et al. 2015, ApJL, 800, L19
Bouquin, A. Y. K., Gil de Paz, A., Muñoz-Mateos, J. C., et al. 2018, ApJS, 234, 18
Brammer, G. B., Whitaker, K. E., van Dokkum, P. G., et al. 2009, ApJL, 706, L173
Braun, R., Walterbos, R. A. M., Kennicutt, R. C., Jr., & Tacconi, L. J. 1994, ApJ, 420, 558
Brinchmann, J., & Ellis, R. S. 2000, ApJL, 536, L77
Buson, L. M., Bertola, F., Bressan, A., Burstein, D., & Cappellari, M. 2004, A&A, 423, 965
Caglar, T., Burtscher, L., Brandl, B., et al. 2020, A&A, 634, A114
Carollo, C. M., & Danziger, I. J. 1994, MNRAS, 270, 743
Carollo, C. M., Stiavelli, M., de Zeeuw, P. T., & Mack, J. 1997, AJ, 114, 2366
Carpenter, B., Gelman, A., Hoffman, M. D., et al. 2017, Journal of Statistical Software, 76, 1
Catalán-Torrecilla, C., Gil de Paz, A., Castillo-Morales, A., et al. 2015, A&A, 584, A87
Cattaneo, A., Dekel, A., Faber, S. M., & Guiderdoni, B. 2008, MNRAS, 389, 567
Cattaneo, A., Faber, S. M., Binney, J., et al. 2009, Natur, 460, 213
Chang, Y.-Y., van der Wel, A., da Cunha, E., & Rix, H.-W. 2015, ApJS, 219, 8
Chen, C.-T. J., Hickox, R. C., Alberts, S., et al. 2013, ApJ, 773, 3
Chester, C., & Roberts, M. S. 1964, AJ, 69, 635
Cisternas, M., Gadotti, D. A., Knapen, J. H., et al. 2013, ApJ, 776, 50
Code, A. D., & Welch, G. A. 1979, ApJ, 228, 95
Conselice, C. J., Gallagher, J. S., Calzetti, D., Homeier, N., & Kinney, A. 2000, AJ, 119, 79
Courteau, S., de Jong, R. S., & Broeils, A. H. 1996, ApJL, 457, L73
Cowie, L. L., Songaila, A., Hu, E. M., & Cohen, J. G. 1996, AJ, 112, 839
Croton, D. J., Springel, D. J., White, V., et al. 2006, MNRAS, 365, 11
Dale, D. A., Cohen, S. A., Johnson, L. C., et al. 2009, ApJ, 703, 517
Davis, B. L., Graham, A. W., & Cameron, E. 2018, ApJ, 869, 113
Davis, B. L., Graham, A. W., & Seigar, M. S. 2017, MNRAS, 471, 2187
Davis, T. A., Greene, J. E., Ma, C.-P., et al. 2019, MNRAS, 486, 1404
de la Rosa, I. G., La Barbera, F., Ferreras, I., et al. 2016, MNRAS, 457, 1916
de La Rosa, I. G., La Barbera, F., Ferreras, I., & de Carvalho, R. R. 2011, MNRAS, 418, L74
Di Matteo, T., Springel, V., & Hernquist, L. 2005, Natur, 433, 604
Dressler, A. 1989, in IAU Symp. 134, Active Galactic Nuclei, ed. D. E. Osterbrock & J. S. Miller (Dordrecht: Kluwer), 217
Driver, S. P., Popescu, C. C., Tuffs, R. J., et al. 2008, ApJL, 678, L101
Dullo, B. T. 2014, in ASP Conf. Ser. 480, Central Stellar Mass Deficits in the Bulges of Local Lenticular Galaxies, ed. M. S. Seigar & P. Treuthardt (San Francisco, CA: ASP), 75
Dullo, B. T. 2019, ApJ, 886, 80
Dullo, B. T., Chamorro-Cazorla, M., Gil de Paz, A., et al. 2019, ApJ, 871, 9
Dullo, B. T., Gil de Paz, A., & Knapen, J. H. 2020, ApJ, in press
Dullo, B. T., & Graham, A. W. 2012, ApJ, 755, 163
Dullo, B. T., & Graham, A. W. 2013, ApJ, 768, 36
Dullo, B. T., & Graham, A. W. 2014, MNRAS, 444, 2700
Dullo, B. T., & Graham, A. W. 2015, ApJ, 798, 55
Dullo, B. T., Graham, A. W., & Knapen, J. H. 2017, MNRAS, 471, 2321
Dullo, B. T., Knapen, J. H., Williams, D. R. A., et al. 2018, MNRAS, 475, 4670
Dullo, B. T., Martínez-Lombilla, C., & Knapen, J. H. 2016, MNRAS, 462, 3800
Ebisuzaki, T., Makino, J., & Okumura, S. K. 1991, Natur, 354, 212
Elmegreen, D. M., Chromey, F. R., McGrath, E. J., & Ostenson, J. M. 2002, AJ, 123, 1381
Erwin, P., Saglia, R. P., Fabricius, M., et al. 2015, MNRAS, 446, 4039
Eskridge, P. B., & Pogge, R. W. 1997, ApJ, 486, 259
Faber, S. M., Tremaine, S., Ajhar, E. A., et al. 1997, AJ, 114, 1771
Fabian, A. C. 1999, MNRAS, 308, L39
Feigelson, E. D., & Babu, G. J. 1992, ApJ, 397, 55
Ferrarese, L., & Ford, H. 2005, SSRv, 116, 523
Ferrarese, L., & Merritt, D. 2000, ApJL, 539, L9
Filippenko, A. V., & Ho, L. C. 2003, ApJL, 588, L13
Fisher, D. B., & Drory, N. 2008, AJ, 136, 773
Gadotti, D. A. 2009, MNRAS, 393, 1531
García-Burillo, S., Combes, F., Hunt, L. K., et al. 2003, A&A, 407, 485
Gebhardt, K., Bender, R., Bower, G., et al. 2000, ApJL, 539, L13
Gelman, A., Carlin, J. B., Stern, H. S., et al. 2013, Bayesian Data Analysis (London: Chapman and Hall)
Ghosh, K. K., Swartz, D. A., Tennant, A. F., Wu, K., & Saripalli, L. 2005, ApJ, 623, 815
Gil de Paz, A., Boissier, S., Madore, B. F., et al. 2007, ApJS, 173, 185
Gourgoulhon, E., Chamaraux, P., & Fouque, P. 1992, A&A, 255, 69
Graham, A. W. 2012, ApJ, 746, 113
Graham, A. W. 2013, in Planets, Stars and Stellar Systems, Extragalactic Astronomy and Cosmology, Vol. 6, ed. T. D. Oswalt & W. C. Keel (Berlin: Springer), 91
Graham, A. W. 2016, in Galactic Bulges, Astrophysics and Space Science Library, Vol. 418, ed. E. Laurikainen, R. Peletier, & D. Gadotti (Berlin: Springer), 263
Graham, A. W., Dullo, B. T., & Savorgnan, G. A. D. 2015, ApJ, 804, 32
Graham, A. W., Erwin, P., Caon, N., & Trujillo, I. 2001, ApJL, 563, L11
Graham, A. W., Erwin, P., Trujillo, I., & Asensio Ramos, A. 2003, AJ, 125, 2951
Graham, A. W., & Scott, N. 2013, ApJ, 764, 151
Graham, A. W., & Worley, C. C. 2008, MNRAS, 388, 1708
Greene, J. E., Ho, L. C., & Barth, A. J. 2008, ApJ, 688, 159
Greene, J. E., Peng, C. Y., Kim, M., et al. 2010, ApJ, 721, 26
Greene, J. E., Seth, A., Kim, M., et al. 2016, ApJL, 826, L32
Gültekin, K., Richstone, D. O., Gebhardt, K., et al. 2009, ApJ, 698, 198
Habouzit, M., Genel, S., Somerville, R. S., et al. 2019, MNRAS, 484, 4413
Hägele, G. F., Díaz, Á I., Cardaci, M. V., Terlevich, E., & Terlevich, R. 2010, MNRAS, 402, 1005
Heckman, T. M., Kauffmann, G., Brinchmann, J., et al. 2004, ApJ, 613, 109
Hilbe, J. M., de Souza, R. S., & Ishida, E. E. O. 2017, Bayesian Models for Astrophysical Data Using R, JAGS, Python, and Stan (Cambridge: Cambridge University Press)
Hilker, M., & Kissler-Patig, M. 1996, A&A, 314, 357
Ho, L. C. 2005, ApJ, 629, 680
Hopkins, P. F., Cox, T. J., Dutta, S. N., et al. 2009a, ApJS, 181, 135
Hopkins, P. F., & Hernquist, L. 2009, ApJ, 694, 599
Hopkins, P. F., Hernquist, L., Cox, T. J., et al. 2006, ApJS, 163, 1
Hopkins, P. F., Lauer, T. R., Cox, T. J., Hernquist, L., & Kormendy, J. 2009b, ApJS, 181, 486
Hopkins, P. F., & Quataert, E. 2010, MNRAS, 407, 1529
Jahnke, K., & Macciò, A. V. 2011, ApJ, 734, 92
Jardel, J. R., Gebhardt, K., Shen, J., et al. 2011, ApJ, 739, 21
Jeong, H., Yi, S. K., Bureau, M., et al. 2009, MNRAS, 398, 2028
Józsa, G. I. G., Oosterloo, T. A., Morganti, R., Klein, U., & Erben, T. 2009, A&A, 494, 489
Karataeva, G. M., Drozdovsky, I. O., Hagen-Thorn, V. A., et al. 2004, AJ, 127, 789

Kauffmann, G., Heckman, T. M., Tremonti, C., et al. 2003a, MNRAS, 346, 1055
Kauffmann, G., Heckman, T. M., White, S. D. M., et al. 2003b, MNRAS, 341, 54
Kaviraj, S. 2014a, MNRAS, 440, 2944
Kaviraj, S. 2014b, MNRAS, 437, L41
Kelly, B. C. 2007, ApJ, 665, 1489
Kennedy, R., Bamford, S. P., Häußler, B., et al. 2016, MNRAS, 460, 3458
Khochfar, S., & Burkert, A. 2001, ApJ, 561, 517
Khochfar, S., & Silk, J. 2009, MNRAS, 397, 506
Kim, D.-W., Guhathakurta, P., van Gorkom, J. H., Jura, M., & Knapp, G. R. 1988, ApJ, 330, 684
King, A. 2003, ApJL, 596, L27
Knapen, J. H., Cisternas, M., & Querejeta, M. 2015, MNRAS, 454, 1742
Knapen, J. H., Erroz-Ferrer, S., Roa, J., et al. 2014, A&A, 569, A91
Kormendy, J. 1982, ApJ, 257, 75
Kormendy, J. 1993, in IAU Symp. 153, Galactic Bulges, ed. H. Dejonghe & H. J. Habing (Dordrecht: Kluwer), 209
Kormendy, J. 2013, in Secular Evolution in Disk Galaxies, ed. J. Falcón-Barroso & J. H. Knapen (Cambridge: Cambridge University Press), 1
Kormendy, J., & Ho, L. C. 2013, ARA&A, 51, 511
Kormendy, J., & Kennicutt, R. C., Jr. 2004, ARA&A, 42, 603
Kormendy, J., & Richstone, D. 1995, ARA&A, 33, 581
Krajnović, D., Cappellari, M., & McDermid, R. M. 2018, MNRAS, 473, 5237
Krajnović, D., Ural, U., Kuntschner, H., et al. 2020, A&A, 635, A129
Lauer, T. R., Faber, S. M., Richstone, D., et al. 2007, ApJ, 662, 808
Laurikainen, E., Salo, H., Buta, R., & Knapen, J. H. 2007, MNRAS, 381, 401
Lee, J. C., Gil de Paz, A., Kennicutt, R. C., Jr., et al. 2011, ApJS, 192, 6
Leitner, S. N. 2012, ApJ, 745, 149
Leonardi, A. J., & Worthey, G. 2000, ApJ, 534, 650
Lutz, D., Sturm, E., Tacconi, L. J., et al. 2008, ApJ, 684, 853
Madau, P., & Dickinson, M. 2014, ARA&A, 52, 415
Magorrian, J., Tremaine, S., Richstone, D., et al. 1998, AJ, 115, 2285
Makarov, D., Prugniel, P., Terekhova, N., Courtois, H., & Vauglin, I. 2014, A&A, 570, A13
Malin, D., & Hadley, B. 1997, PASA, 14, 52
Man, A. W. S., Toft, S., Zirm, A. W., Wuyts, S., & van der Wel, A. 2012, ApJ, 744, 85
Marconi, A., & Hunt, L. K. 2003, ApJL, 589, L21
Marleau, F. R., Noriega-Crespo, A., Misselt, K. A., et al. 2006, ApJ, 646, 929
Martin, D. C., Fanson, J., Schiminovich, D., et al. 2005, ApJL, 619, L1
Martin, G., Kaviraj, S., Volonteri, M., et al. 2018a, MNRAS, arXiv:1801.09699
Martin, G., Kaviraj, S., Volonteri, M., et al. 2018b, MNRAS, 476, 2801
Martín-Navarro, I., Brodie, J. P., Romanowsky, A. J., Ruiz-Lara, T., & van de Ven, G. 2018, Natur, 553, 307
McConnell, N. J., & Ma, C.-P. 2013, ApJ, 764, 184
McDermid, R. M., Alatalo, K., Blitz, L., et al. 2015, MNRAS, 448, 3484
McLure, R. J., & Dunlop, J. S. 2002, MNRAS, 331, 795
Meidt, S. E., Schinnerer, E., Knapen, J. H., et al. 2012, ApJ, 744, 17
Merloni, A., Rudnick, G., & Di Matteo, T. 2004, MNRAS, 354, L37
Merritt, D. 2006, ApJ, 648, 976
Möllenhoff, C. 2004, A&A, 415, 63
Möllenhoff, C., Popescu, C. C., & Tuffs, R. J. 2006, A&A, 456, 941
Morrissey, P., Conrow, T., Barlow, T. A., et al. 2007, ApJS, 173, 682
Mulder, P. S., van Driel, W., & Braine, J. 1995, A&A, 300, 687
Mundy, C. J., Conselice, C. J., Duncan, K. J., et al. 2017, MNRAS, 470, 3507
Muñoz-Mateos, J. C., Sheth, K., Regan, M., et al. 2015, ApJS, 219, 3
Murray, N., Quataert, E., & Thompson, T. A. 2005, ApJ, 618, 569
Naab, T., Jesseit, R., & Burkert, A. 2006a, MNRAS, 372, 839
Naab, T., Khochfar, S., & Burkert, A. 2006b, ApJL, 636, L81
Nemmen, R. S., Georganopoulos, M., Guiriec, S., et al. 2012, Sci, 338, 1445
Netzer, H. 2009, MNRAS, 399, 1907
Nguyen, D. D., Seth, A. C., Neumayer, N., et al. 2019, ApJ, 872, 104
Nolan, L. A., Raychaudhury, S., & Kabán, A. 2007, MNRAS, 375, 381
Nowak, N., Thomas, J., Erwin, P., et al. 2010, MNRAS, 403, 646
O'Connell, R. W. 1999, ARA&A, 37, 603
O'Dea, C. P., Baum, S. A., Privon, G., et al. 2008, ApJ, 681, 1035
Pannella, M., Carilli, C. L., Daddi, E., et al. 2009, ApJL, 698, L116
Paturel, G., Petit, C., Prugniel, P., et al. 2003, A&A, 412, 45
Peng, C. Y. 2007, ApJ, 671, 1098
Pérez-González, P. G., Rieke, G. H., Villar, V., et al. 2008, ApJ, 675, 234
Peterson, B. M. 1993, PASP, 105, 247
Peterson, B. M., Bentz, M. C., Desroches, L.-B., et al. 2005, ApJ, 632, 799
Rampazzo, R., Marino, A., Tantalo, R., et al. 2007, MNRAS, 381, 245
Richstone, D., Ajhar, E. A., Bender, R., et al. 1998, Natur, 395, A14
Rodriguez-Gomez, V., Genel, S., Vogelsberger, M., et al. 2015, MNRAS, 449, 49
Rodriguez-Gomez, V., Pillepich, A., Sales, L. V., et al. 2016, MNRAS, 458, 2371
Rosario, D. J., Santini, D. J., Lutz, P., et al. 2012, A&A, 545, A45
Russell, H. R., McNamara, B. R., Edge, A. C., et al. 2014, ApJ, 784, 78
Russell, H. R., McNamara, B. R., Fabian, A. C., et al. 2019, MNRAS, 490, 3025
Saglia, R. P., Opitsch, M., Erwin, P., et al. 2016, ApJ, 818, 47
Sahu, N., Graham, A. W., & Davis, B. L. 2019a, ApJ, 876, 155
Sahu, N., Graham, A. W., & Davis, B. L. 2019b, ApJ, 887, 10
Salo, H., Laurikainen, E., Laine, J., et al. 2015, ApJS, 219, 4
Salomé, P., & Combes, F. 2003, A&A, 412, 657
Sandage, A. 1961, The Hubble Atlas of Galaxies (Washington, DC: Carnegie Institution)
Sanders, D. B., Soifer, B. T., Elias, J. H., et al. 1988, ApJ, 325, 74
Savorgnan, G. A. D., & Graham, A. W. 2016, ApJS, 222, 10
Savorgnan, G. A. D., Graham, A. W., Marconi, A., & Sani, E. 2016, ApJ, 817, 21
Schawinski, K., Khochfar, S., Kaviraj, S., et al. 2006, Natur, 442, 888
Schawinski, K., Urry, C. M., Simmons, B. D., et al. 2014, MNRAS, 440, 889
Schawinski, K., Urry, C. M., Virani, S., et al. 2010, ApJ, 711, 284
Schechter, P. L., & Gunn, J. E. 1978, AJ, 83, 1360
Schinnerer, E., & Scoville, N. 2002, ApJL, 577, L103
Schlegel, D. J., Finkbeiner, D. P., & Davis, M. 1998, ApJ, 500, 525
Schweizer, F. 1987, in IAU Symp. 127, Structure and Dynamics of Elliptical Galaxies, ed. P. T. de Zeeuw (Dordrecht: Reidel), 109
Segers, M. C., Schaye, J., Bower, R. G., et al. 2016, MNRAS, 461, L102
Sérsic, J. L. 1968, Atlas de Galaxias Australes (Cordoba: Observatorio Astronomico)
Shane, W. W. 1980, A&A, 82, 314
Shankar, F., Weinberg, D. H., & Miralda-Escudé, J. 2009, ApJ, 690, 20
Sheth, K., Regan, M., Hinz, J. L., et al. 2010, PASP, 122, 1397
Sijacki, D., Vogelsberger, M., Genel, S., et al. 2015, MNRAS, 452, 575
Sil'chenko, O. K., Moiseev, A. V., Afanasiev, V. L., Chavushyan, V. H., & Valdes, J. R. 2003, ApJ, 591, 185
Silk, J., & Rees, M. J. 1998, A&A, 331, L1
Simmons, B. D., Lintott, C., Schawinski, K., et al. 2013, MNRAS, 429, 2199
Simmons, B. D., Smethurst, R. J., & Lintott, C. 2017, MNRAS, 470, 1559
Smith, D. A., Neff, S. G., Bothun, G. D., et al. 1996, ApJL, 473, L21
Smith, R. J., & Edge, A. C. 2017, MNRAS, 471, L66
Spitler, L. R., Forbes, D. A., & Beasley, M. A. 2008, MNRAS, 389, 1150
Springel, V., White, S. D. M., Jenkins, A., et al. 2005, Natur, 435, 629
Steinmetz, M., & Navarro, J. F. 2002, NewA, 7, 155
Struve, C., Oosterloo, T., Sancisi, R., Morganti, R., & Emonts, B. H. C. 2010, A&A, 523, A75
Terrazas, B. A., Bell, E. F., Woo, J., & Henriques, B. M. B. 2017, ApJ, 844, 170
Thomas, D., Maraston, C., Bender, R., & Mendes de Oliveira, C. 2005, ApJ, 621, 673
Thomas, D., Maraston, C., Schawinski, K., Sarzi, M., & Silk, J. 2010, MNRAS, 404, 1775
Toba, Y., Nagao, T., Wang, W.-H., et al. 2017, ApJ, 840, 21
Tonini, C., Mutch, S. J., Croton, D. J., & Wyithe, J. S. B. 2016, MNRAS, 459, 4109
Toomre, A., & Toomre, J. 1972, ApJ, 178, 623
Treuthardt, P., Buta, R., Salo, H., & Laurikainen, E. 2007, AJ, 134, 1195
van den Bosch, R. C. E. 2016, ApJ, 831, 134
van Driel, W., & Buta, R. 1993, PASJ, 45, L47
van Son, L. A. C., Barber, C., Bahé, Y. M., et al. 2019, MNRAS, 485, 396
Visvanathan, N., & Griersmith, D. 1977, A&A, 59, 317
Watkins, A. E., Mihos, J. C., Harding, P., & Feldmeier, J. J. 2014, ApJ, 791, 38
Watson, D. M., Guptill, M. T., & Buchholz, L. M. 1994, ApJL, 420, L21
Wehner, E. H., Gallagher, J. S., Papaderos, P., Fritze-von Alvensleben, U., & Westfall, K. B. 2006, MNRAS, 371, 1047
Wehner, E. H., & Gallagher, J. S., III 2005, ApJL, 618, L21
Weinberger, R., Springel, V., Hernquist, L., et al. 2017, MNRAS, 465, 3291
White, S. D. M., & Rees, M. J. 1978, MNRAS, 183, 341
Wild, V., Heckman, T., & Charlot, S. 2010, MNRAS, 405, 933
Yang, G., Chen, C.-T. J., Vito, F., et al. 2017, ApJ, 842, 72
Yi, S. K., Lee, J., Sheen, Y.-K., et al. 2011, ApJS, 195, 22
Young, L. M., Bureau, M., Davis, T. A., et al. 2011, MNRAS, 414, 940
Zubovas, K., & King, A. R. 2012, MNRAS, 426, 2751